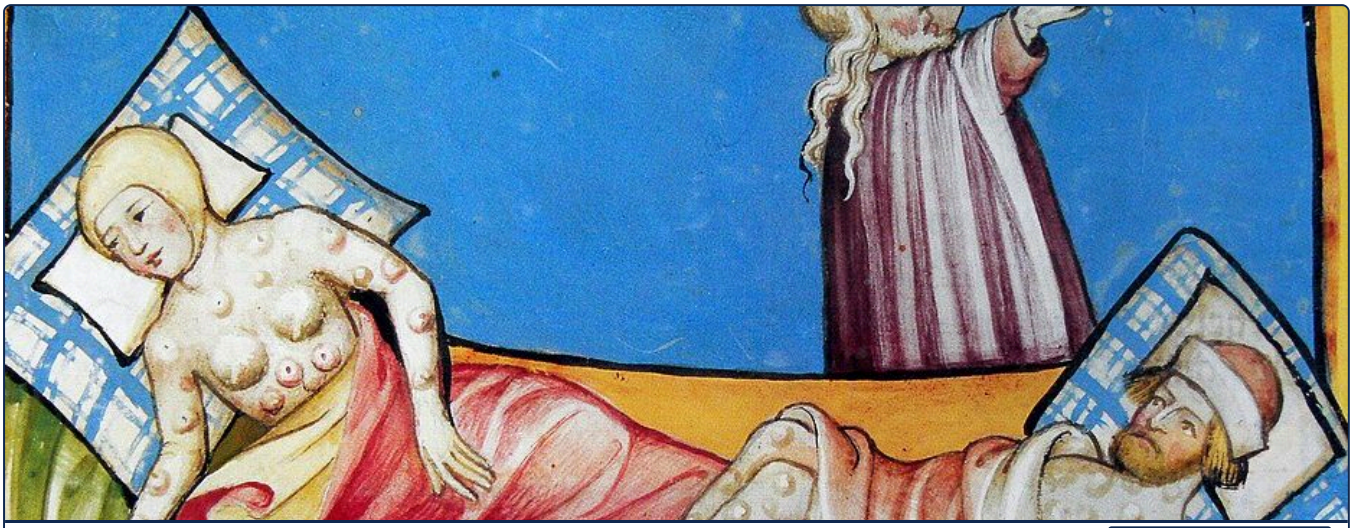
PUPIL NAME: _____ TARGET: _____ CLASS: _____

KEY TOPIC 1 • C1250–C1500 CHRONOLOGICAL ENQUIRY SEQUENCE

Medicine in Medieval Britain, c1250–c1500

c1250–c1500: Ideas on Causes, Prevention, Treatment, Medical Care & The Black Death



HISTORICAL PRIMARY RECORD • C.1350 **MS 13076 • FOLIO 24V**

Omne Bonum: Clerics Ministering to Plague Victims with Buboes

James le Palmer (fl. 1360–1375) • Fourteenth-century illuminated manuscript showing two plague sufferers bearing bubonic sores in bed while priests administer last rites and prayer.

PRIMARY VISUAL EVIDENCE **THE BRITISH LIBRARY MANUSCRIPT COLLECTION**

KEY TOPIC 1 SPECIFICATION FRAMEWORK (EDEXCEL 1HI0/11)	LESSON DONE	QUIZ [10]
KT1.1: Supernatural & Religious Explanations of Disease (c1250–c1500) *Why did the medieval Catholic Church hold an unshakeable monopoly over beliefs about the causes of disease?* **SPEC:** Supernatural and religious explanations of the cause of disease and illness • Rational explanations: the Theory of the Four Humours and the miasma theory • The continuing influence in England of Galen	☐	☐
KT1.2: Rational Explanations: Hippocrates, Galen & The Four Humours (c1250–c1500) *How did the Theory of the Four Humours provide a logical, complete explanation for every illness in medieval England?* **SPEC:** Approaches to prevention and treatment, and their connection with ideas about disease and illness: religious actions, bloodletting and purging, purifying the air	☐	☐
KT1.3: Approaches to Prevention & Treatment: Rituals, Bleeding & Purging (c1250–c1500) *Why did standard medieval medical treatments frequently endanger the lives of the patients they were meant to cure?* **SPEC:** Medical training and traditional approaches to treatment and care for the sick: role of the physician, apothecary and barber surgeon; hospitals, care within community and home, herbal remedies	☐	☐
KT1.4: Medical Care Providers & Monastic Hospitals: 'Care Not Cure' (c1250–c1500) *How did social hierarchy and the Catholic Church shape who provided medical care in medieval England?* **SPEC:** Dealing with the Black Death, 1348–49: approaches to treatment and attempts to prevent its spread	☐	☐
KT1.5: Case Study: Dealing with the Black Death (1348–1349) *Why were medieval communities, doctors, and civic authorities completely powerless to stop the Black Death?* **SPEC:** Thematic synthesis: ideas on causes, prevention, treatment, and medical care c1250–c1500; exam question practice	☐	☐

Living Timeline • Part 1: Medieval Ideas & Medical Hierarchy (c1250–c1350)

Living Timeline Dual-Coding Protocol: Complete each milestone sketchpad using dual-coding (combining visual storytelling symbols, causal arrows, and dates) to master the chronological spine across the centre-fold.

c. 1250 • The Church & The Monopoly on Medical Learning

Key Topic 1.1

The Catholic Church controls all English universities (Oxford, Cambridge) and hospital foundations. Monks hand-copy medical manuscripts in scriptoria, preserving Galen's anatomical writings because his teleological view—that the body had a single divine Creator—aligned with Christian dogma.

1277 • The Imprisonment of Roger Bacon

Key Topic 1.1

Franciscan friar and Oxford scholar Roger Bacon is imprisoned by church leaders for heresy after writing that physicians should conduct original scientific experiments rather than blindly copying Galen. This imprisonment effectively halts independent anatomical inquiry for over two centuries.

c. 1300 • The Diagnostic Tools: Matula Flasks & The Zodiac Man

Key Topic 1.2

Physicians practice uroscopy using glass matula flasks, comparing urine colour, density, and smell against 20-shade charts to diagnose humoral imbalance. They carry Vademecum pocketbooks containing the Zodiac Man to determine safe planetary alignments for phlebotomy (bloodletting).

c. 1320 • The Medical Hierarchy: Physicians, Barber-Surgeons & Apothecaries

Key Topic 1.4

A rigid division of medical labour establishes in England. Fewer than 100 elite university-trained physicians diagnose wealthy nobility. Town apothecaries mix theriac and herbal compounds, while lower-status barber-surgeons perform tooth-pulling, wound-cautery, and bloodletting.

Living Timeline Dual-Coding Protocol: Complete each milestone sketchpad using dual-coding (combining visual storytelling symbols, causal arrows, and dates) to master the chronological spine across the centre-fold.

c. 1350 • Monastic Hospitals: The Era of 'Care Not Cure'

Key Topic 1.4

Over 1,200 hospitals run by monks and nuns operate in England (e.g. St Bartholomew's, 1123). Patients receive clean bedding, warmth, food, and continuous prayer facing a chapel altar. Zero surgery or medical cure is attempted; lepers and plague victims are strictly barred.

1345 • The Great Planetary Conjunction in Aquarius

Key Topic 1.5

In March 1345, Mars, Jupiter, and Saturn align in Aquarius. University of Paris physicians state this celestial conjunction drew up poisonous vapours from the earth, corrupting the air with deadly miasma, preparing Europe for the Black Death.

1348 • The Black Death Reaches Melcombe Regis, Dorset

Key Topic 1.5

In June 1348, the pestilence arrives on trade ships carrying black rats and fleas. The dual epidemic—bubonic plague (flea bites, agonizing buboes) and pneumonic plague (airborne coughing droplets)—kills 30–50% of England's population within 18 months.

1349 • King Edward III's Cleanliness Order & East Smithfield Mass Pits

Key Topic 1.5

Alarmed by rotting waste and stench, Edward III orders London streets cleared of dung to remove miasma. Churchyards overflow; emergency mass burial trenches are dug outside city walls at East Smithfield to bury hundreds of corpses daily.

Why did the Catholic Church dominate medieval explanations of disease?

Core Specification Focus: Supernatural & religious causes; Catholic Church influence; astrology & planetary conjunctions.

1 c1250 · How did church scriptoria prevent new medical ideas from spreading? [1.1]

- **Hand-copying Latin manuscripts by candle-light**
- **Training university priests in Church doctrine**
- **Punishing medical heretics like Roger Bacon**

2 c1280 · Why was leprosy viewed as visible proof of divine judgment? [1.2]

- **Sending sickness to punish human sin**
- **Exiling lepers to isolated Lazar houses**
- **Ringing warning bells in dirt lanes**

3 1345 · How did the Church reconcile astrology with Christian doctrine? [1.3]

- **Consulting parchment zodiac almanac wheels**
- **Aligning Mars, Jupiter, and Saturn in Aquarius**
- **Blaming corrupted night air for fever**

4 c1350 · Why did sick people turn to pilgrimage shrines rather than doctors? [1.4]

- **Lighting beeswax candles before parish shrines**
- **Kissing holy bone relics of saints**
- **Selling paper indulgences to wealthy barons**

5 c1400 · Why did church enforcement of Galen prevent medical progress? [1.5]

- **Treating Galen's pagan texts as sacred truth**
- **Banning human dissections in English towns**
- **Threatening curious scholars with church excommunication**

If you develop a fever in 1300, church logic dictates you definitely committed a sin last Tuesday. Time to buy a pilgrimage badge!

Question 4: Explain Why [12 marks • 15 mins]

EXTENDED WRITING ASSESSMENT

Explain why the Catholic Church had such a major influence on ideas about the causes of disease in the period c1250–c1500. [12 marks]

Stimulus: **Monastic scriptoria and book-copying** | **The imprisonment of Roger Bacon (1277)** *(You must also use information of your own)*

ACCESSIBLE WRITING FRAMES:

One main reason was that... • For example, ... • This was important because... • Another key factor was... • In particular, ... • Overall, the most decisive reason was... because...

KEY VOCABULARY BANK:

divine retribution • biblical leprosy • Lazar houses • teleology • heresy • Roger Bacon (1277) • monastic scriptoria • 1345 planetary conjunction • astrological almanacs

Write your response in the space provided below:

(Response continues on facing page)

TIMELINE MISSION Turn to Pages 2–3 (Key Topic 1.1). In the drawing box, sketch a monk copying Galen's medical book in a scriptorium. Add two speech bubbles explaining why the Church banned new medical experiments and imprisoned Roger Bacon in 1277. **[ ] Done**

Continue your response to Question 4 below:

• **QUESTION 3: EXPLAIN ONE SIMILARITY [4 MARKS]** SCORE: [/ 4] • 5 MINS

Explain one way in which ideas about the cause of disease in the Medieval period (c1250–c1500) were similar to ideas in the Renaissance period (c1500–c1700). [4 marks]

Identify 1 valid similarity, then explain with specific historical detail from both named eras.

Why did the Theory of the Four Humours dominate rational medicine for centuries?

Core Specification Focus: Rational explanations: Four Humours (Hippocrates), Theory of Opposites (Galen), Miasma; diagnostic tools.

1 c460 BC • How did Hippocrates separate medicine from superstition? [2.1]

• Rejecting anger of gods at the bedside	
• Writing case charts on fever symptoms	
• Swearing ethical oath to protect patients	

2 c1250 • Why did humoral theory seem logical to medieval people? [2.2]

• Balancing blood, phlegm, and two biles	
• Linking fluids to four earthly elements	
• Associating cold damp seasons with winter coughs	

3 c1280 • How did the Theory of Opposites provide practical treatment rules? [2.3]

• Cooling burning summer fever with cold cucumber	
• Treating excess phlegm with dry hot pepper	
• Dissecting Barbary apes to map organs	

4 c1300 • Why did uroscopy give physicians great diagnostic authority? [2.4]

• Holding glass flasks up to window sunlight	
• Comparing urine colours against twenty-shade wheels	
• Sniffing and tasting samples for sweetness	

5 c1350 • Why did rational physicians link bad smells directly to epidemics? [2.5]

• Fearing stench from open roadside cesspools	
• Smelling rotting town refuse heaps	
• Believing toxic smells unbalanced bodily fluids	

Question 4: Explain Why [12 marks • 15 mins]

EXTENDED WRITING ASSESSMENT

Explain why the Theory of the Four Humours remained the dominant rational explanation of disease throughout the period c1250–c1500. [12 marks]

Stimulus: Galen's Theory of Opposites | The diagnostic matula (uroscopy) *(You must also use information of your own)*

ACCESSIBLE WRITING FRAMES:

One main reason was that... • For example, ... • This was important because... • Another key factor was... • In particular, ... • Overall, the most decisive reason was... because...

KEY VOCABULARY BANK:

Four Humours • blood, phlegm, yellow bile, black bile • Theory of Opposites • Galen • clinical observation • Hippocratic Oath • matula • uroscopy wheel • Zodiac Man • miasma

Write your response in the space provided below:

(Response continues on facing page)

TIMELINE MISSION Turn to Pages 2–3 (Key Topic 1.2). In the drawing box, sketch a four-part diagram of the Four Humours (Blood, Phlegm, Yellow Bile, Black Bile) showing their seasons and qualities. Annotate Galen's Theory of Opposites alongside a flask of urine. [ ] Done

Sniffing urine, tasting urine, holding urine up to the window. 14th-century diagnostics was 90% liquid inspection and 10% guessing.

Continue your response to Question 4 below:

• **QUESTION 3: EXPLAIN ONE DIFFERENCE [4 MARKS]** SCORE: [/ 4] • 5 MINS

Explain one way in which ideas about the cause of disease in the Medieval period (c1250–c1500) were different from ideas in the Modern period (c1900–present). [4 marks]

Identify 1 valid difference, then explain with specific historical detail from both named eras.

How did medieval people attempt to treat and prevent disease, and why did rational and religious methods coexist?

Core Specification Focus: Approaches to treatment and prevention; religious & supernatural treatments; humoral treatments (phlebotomy, purging); herbal remedies; regimens.

1 c1250 · Why did ordinary people consider prayer more vital than medicine? [3.1]

- **Chanting Latin prayers beside the sickbed**
- **Fasting on bread and well water**
- **Whipping bare backs to beg God for mercy**

2 c1300 · Why was bloodletting the most universal physical treatment? [3.2]

- **Slicing vein walls with iron fleams**
- **Applying fat swamp leeches to temples**
- **Catching warm dark blood in pewter bowls**

3 c1320 · How was purging believed to restore bodily equilibrium? [3.3]

- **Swallowing poisonous black hellebore seeds**
- **Pumping warm mallow tea through boar-bladder clysters**
- **Emptying bowels to balance bodily humours**

4 c1350 · Why did herbal recipes remain popular across all social classes? [3.4]

- **Grinding crushed mint and sage in stone mortars**
- **Mixing seventy ingredients into treacle theriac**
- **Brewing bitter willow bark teas for fever**

5 c1400 · How did wealthy barons attempt to prevent sickness before it struck? [3.5]

- **Following private Latin health rulebooks**
- **Burning sweet lavender sticks indoors**
- **Carrying fragrant posy flowers through smelly streets**

Bleeding, purging, boiling vinegar, and swallowing crushed emeralds. If the disease didn't kill you, the apothecary's bill definitely would.

Question 4: Explain Why [12 marks • 15 mins]

EXTENDED WRITING ASSESSMENT

Explain why bloodletting and purging remained the most common medical treatments throughout the period c1250–c1500. [12 marks]

Stimulus: **Phlebotomy (bloodletting by fleam or leeches)** **The Regimen Sanitatis** *(You must also use information of your own)*

ACCESSIBLE WRITING FRAMES:

One main reason was that... • For example, ... • This was important because... • Another key factor was... • In particular, ... • Overall, the most decisive reason was... because...

KEY VOCABULARY BANK:

phlebotomy • fleam • leeches • cupping • purging • emetics • clysters • theriac • herbal materia medica • Regimen Sanitatis • flagellants

Write your response in the space provided below:

(Response continues on facing page)

TIMELINE MISSION Turn to Pages 2–3 (Key Topic 1.3). In the drawing box, sketch a barber-surgeon using a fleam and pewter bleeding bowl, alongside a flagellant whip. Label how one treated the body and the other treated the soul. **[ ] Done**

Continue your response to Question 4 below:

• **QUESTION 3: EXPLAIN ONE SIMILARITY [4 MARKS]** **SCORE: [/ 4] • 5 MINS**

Explain one way in which approaches to preventing disease in the Medieval period (c1250–c1500) were similar to approaches in the Renaissance period (c1500–c1700). [4 marks]

Identify 1 valid similarity, then explain with specific historical detail from both named eras.

Who provided medical care in medieval England, and why did hospitals focus on care rather than cure?

Core Specification Focus: Medical care providers: physicians, apothecaries, barber-surgeons; women in the home; hospitals (monastic care, care not cure).

1 c1250 · Why were university physicians accessible only to the ultra-wealthy? [4.1]

• Studying Latin texts for ten years in Oxford	
• Consulting astronomical star charts for baronial clients	
• Charging heavy silver fees without touching patients	

2 c1300 · How did apothecaries provide accessible medicines to townspeople? [4.2]

• Weighing dried poppy roots on brass scales	
• Selling cheap herbal pills to town weavers	
• Learning trade secrets through guild apprenticeships	

3 c1320 · Why were barber-surgeons viewed as lower-status craftsmen? [4.3]

• Pulling rotten teeth with iron pliers	
• Setting broken arm bones with wooden splints	
• Cauterising bleeding flesh wounds with red-hot irons	

4 c1350 · Why did monastic hospitals deliberately avoid medical treatment? [4.4]

• Sheltering bedridden pilgrims in monastic wards	
• Providing warm woollen blankets and fresh vegetable stew	
• Hearing Latin mass through chapel chancel screens	

5 c1400 · Why did the vast majority of sick people rely on female family members? [4.5]

• Gathering wild chamomile in country hedgerows	
• Delivering newborn babies in village cottages	
• Applying warm bread poultices to swollen skin	

Question 4: Explain Why [12 marks • 15 mins]

EXTENDED WRITING ASSESSMENT

Explain why hospital care in medieval England was based on the concept of 'care rather than cure' in the period c1250–c1500. [12 marks]

Stimulus: **Monastic rules and religious charity** | **The exclusion of infectious diseases (lepers and plague victims)** *(You must also use information of your own)*

ACCESSIBLE WRITING FRAMES:

One main reason was that... • For example, ... • This was important because... • Another key factor was... • In particular, ... • Overall, the most decisive reason was... because...

KEY VOCABULARY BANK:

care not cure • monastic hospitals • St Bartholomew's (1123) • infirmarer • university physicians • apothecaries • barber-surgeons • wise women • fleam • Lazar houses

Write your response in the space provided below:

(Response continues on facing page)

TIMELINE MISSION Turn to Pages 2–3 (Key Topic 1.4). In the drawing box, sketch the interior of a monastic hospital ward showing rows of beds facing a chapel altar. Annotate why monks prioritized prayer over medical surgery. **[ ] Done**

Barber-surgeons: short back and sides, pull out a molar, lance a boil, amputate a leg. Multi-tasking at its finest and bloodiest.

Continue your response to Question 4 below:

• **QUESTION 3: EXPLAIN ONE DIFFERENCE [4 MARKS]** **SCORE: [/ 4] • 5 MINS**

Explain one way in which the work of hospitals in the Medieval period (c1250–c1500) was different from hospitals in the nineteenth century. [4 marks]

Identify 1 valid difference, then explain with specific historical detail from both named eras.

Why did medieval society fail to halt the devastating mortality of the Black Death?

Core Specification Focus: Case study: The Black Death, 1348–49; beliefs about causes; approaches to treatment and prevention; local government action.

1 June 1348 • How did trade routes unwittingly bring the plague to England? [5.1]

• Docking trade cogs from Gascony in Dorset	
• Unloading flea-infested wool sacks onto wharves	
• Spreading black rats into crowded timber cottages	

2 Autumn 1348 • Why was the pneumonic strain far deadlier than the bubonic form? [5.2]

• Swelling agonizing egg-sized groin lumps	
• Coughing bloody phlegm across family rooms	
• Killing infected farm labourers within three days	

3 Winter 1348 • Why were medical treatments completely useless against the plague? [5.3]

• Lancing hard neck swellings with hot knives	
• Strapping plucked live hens to dark sores	
• Drinking crushed emerald stones dissolved in vinegar	

4 1349 • How did belief in miasma shape royal prevention orders? [5.4]

• Ordering town bailiffs to clear butchered animal entrails	
• Shovelling horse dung out of unpaved London streets	
• Lighting smoking pitch fires at road corners	

5 1349–1350 • Why did the Black Death trigger a long-term social and economic crisis? [5.5]

• Digging vast emergency trenches at East Smithfield	
• Piling stacked corpses under thin river clay	
• Losing half of England's agricultural workforce	

In 1348, people blamed God, the planets, bad smells, and foreigners. The actual culprits—fleas on black rats—went completely unnoticed.

Question 5/6: Thematic Essay [16 marks + 4 marks SPaG • 25 mins]

EXTENDED WRITING ASSESSMENT

'The main reason why so many people died during the Black Death (1348–49) was because governments had no effective way of preventing disease.' How far do you agree? Explain your answer. [16 marks + 4 marks SPaG]

Stimulus: **King Edward III's cleanliness order (1349)** | **Belief in miasma and supernatural causes** | *(You must also use information of your own)*

ACCESSIBLE WRITING FRAMES:

One main reason was that... • For example, ... • This was important because... • Another key factor was... • In particular, ... • Overall, the most decisive reason was... because...

KEY VOCABULARY BANK:

Black Death (1348) • Melcombe Regis • bubonic plague • pneumonic plague • Yersinia pestis • black rats • Xenopsylla cheopis fleas • buboes • King Edward III • East Smithfield mass graves • miasma

Write your response in the space provided below:

(Response continues on facing page)

TIMELINE MISSION Turn to Pages 2–3 (Key Topic 1.5). In the drawing box, sketch the arrival of a trading cog at Melcombe Regis carrying rats and fleas, alongside an emergency mass burial pit at East Smithfield. **[ ] Done**

Continue your response to Question 5/6 below:

• **QUESTION 3: EXPLAIN ONE SIMILARITY [4 MARKS]** SCORE: [/ 4] • 5 MINS

Explain one way in which responses to the Black Death (1348–49) were similar to responses to the Great Plague of London (1665). [4 marks]

Identify 1 valid similarity, then explain with specific historical detail from both named eras.

KEY TOPIC 1 NOTES & EXAMINATION STRATEGY

PUPIL NAME: _____ TARGET: _____ CURRENT GRADE: _____

ENQUIRY	ASSESSMENT COMPONENT FOCUS	PAGE	SCORE	TEACHER COMMENT & NEXT STEPS
KT1.1	**Q4:** Explain Why Church Dominated Explanations c1250–c1500	p. 5–6	/ 12	
	Q3: Explain One Similarity (Causes: Medieval vs Renaissance)	p. 6	/ 4	
KT1.2	**Q4:** Explain Why Four Humours Remained Dominant c1250–c1500	p. 8–9	/ 12	
	Q3: Explain One Difference (Causes: Medieval vs Modern)	p. 9	/ 4	
KT1.3	**Q4:** Explain Why Treatments Focused on Humours & Repentance	p. 11–12	/ 12	
	Q3: Explain One Similarity (Prevention: Medieval vs Renaissance)	p. 12	/ 4	
KT1.4	**Q4:** Explain Why Hospital Care Was Based on 'Care Not Cure'	p. 14–15	/ 12	
	Q3: Explain One Difference (Hospitals: Medieval vs 19th Century)	p. 15	/ 4	
KT1.5	**Q5/6:** Evaluative Essay: Black Death & Government Prevention	p. 17–18	/ 20	
	Q3: Explain One Similarity (Plague Responses: 1348 vs 1665)	p. 18	/ 4	
TOTAL	**Key Topic 1 Cumulative Examination Assessment Portfolio (10 Tasks)**	p. 4–18	/ 88	**Overall Grade:** _____ Teacher Target: _____

• DIGITAL LEARNING & RETRIEVAL HUB • INTERACTIVE LESSONS & QUIZZES

Interactive Textbook • 10 Questions Per Enquiry

KT1.1 Supernatural	KT1.2 Four Humours	KT1.3 Treatments	KT1.4 Hospitals	KT1.5 Black Death
LESSON HUB	LESSON HUB	LESSON HUB	LESSON HUB	LESSON HUB
PRACTICE QUIZ	PRACTICE QUIZ	PRACTICE QUIZ	PRACTICE QUIZ	PRACTICE QUIZ
Score: [/ 10]	Score: [/ 10]	Score: [/ 10]	Score: [/ 10]	Score: [/ 10]

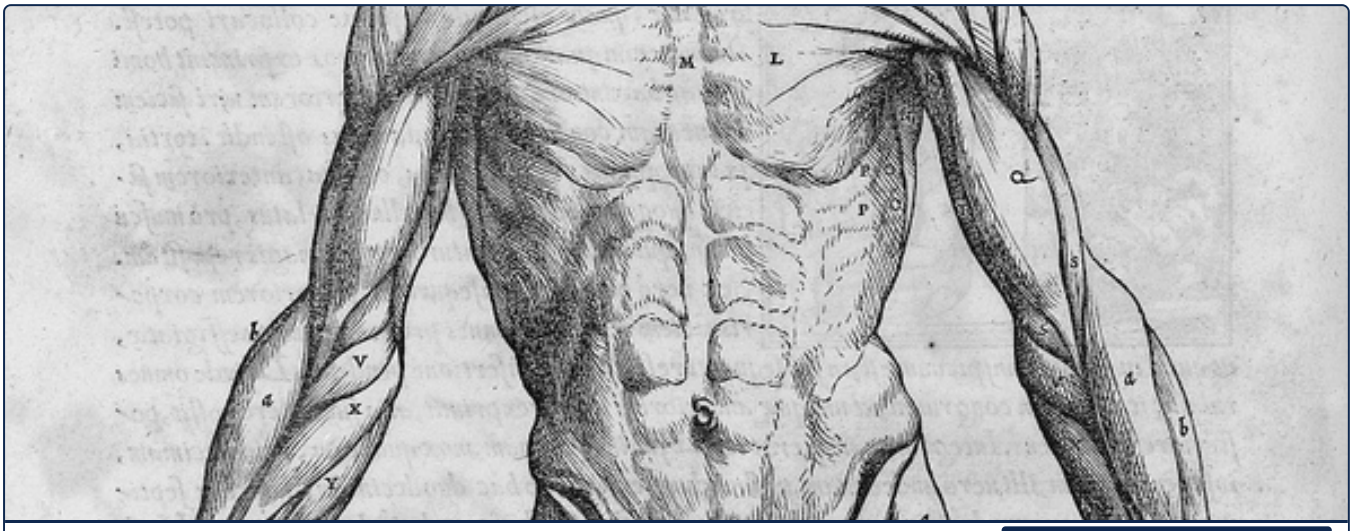
Superstition, humours, and herbal mixtures gave way to observation; retrieval practice will guarantee your GCSE exam success.

PUPIL NAME: _____ TARGET: _____ CLASS: _____

KEY TOPIC 2 • C1500–C1700 CHRONOLOGICAL ENQUIRY SEQUENCE

The Medical Renaissance in Britain, c1500–c1700

c1500–c1700: Humanism, Anatomy, Circulation, Scientific Observation & The Great Plague



HISTORICAL PRIMARY RECORD • 1543 **CONTEMPORARY PRINT ARCHIVE**

Title Page of Andreas Vesalius, De Humani Corporis Fabrica (1543)

Woodcut title page showing Andreas Vesalius conducting an open human dissection surrounded by students, challenging Galenic anatomy directly from empirical observation.

PRIMARY VISUAL EVIDENCE **WELLCOME COLLECTION HISTORICAL ARCHIVE**

KEY TOPIC 2 SPECIFICATION FRAMEWORK (EDEXCEL 1HI0/11)	LESSON DONE	QUIZ [10]
KT2.1: The New Spirit of Enquiry: Humanism, The Printing Press & The Royal Society (c1500–c1700) *How did the invention of printing and the scientific method challenge 1,500 years of medical orthodoxy?* **SPEC:** Continuity and change in explanations of the cause of disease and illness. A scientific approach, including Thomas Sydenham; printing press and the Royal Society	☐	☐
KT2.2: Thomas Sydenham & The Art of Bedside Observation (1676) *How did Thomas Sydenham's focus on direct observation challenge traditional medical practice?* **SPEC:** Continuity and change in approaches to prevention, treatment and care in the community and in hospitals. Improvements in medical training and the work of Vesalius	☐	☐
KT2.3: Andreas Vesalius & The Anatomical Revolution (1543) *Why did Andreas Vesalius succeed in overturning 1,400 years of Galenic anatomical authority?* **SPEC:** Key individual: William Harvey and the discovery of the circulation of the blood	☐	☐
KT2.4: William Harvey & The Circulation of the Blood (1628) *Why did William Harvey's discovery of blood circulation meet fierce medical resistance?* **SPEC:** Dealing with the Great Plague in London (1665): approaches to treatment and attempts to prevent its spread	☐	☐
KT2.5: Continuity in Treatment & The Great Plague of London (1665) *Why did the Great Plague of 1665 reveal the severe limitations of Renaissance medical progress?* **SPEC:** Thematic synthesis: scientific observation, Vesalius, Harvey, and limits of Renaissance medical progress c1500–c1700	☐	☐

Living Timeline • Part 1: Scientific Breakthroughs & Anatomy (c1440–1628)

Living Timeline Dual-Coding Protocol: Complete each milestone sketchpad using dual-coding (combining visual storytelling symbols, causal arrows, and dates) to master the chronological spine across the centre-fold.

c. 1440–1476 • The Printing Press: Gutenberg & William Caxton

Key Topic 2.1

Johannes Gutenberg invents movable metal type in Mainz; William Caxton establishes England's first press at Westminster in 1476. Medical texts can now be mass-produced with identical diagrams, shattering the Catholic Church's clerical monopoly on manuscript copying.

1543 • Andreas Vesalius Publishes De Humani Corporis Fabrica

Key Topic 2.3

At the University of Padua, professor Andreas Vesalius carries out human dissections personally. The Fabric of the Human Body features precise woodcuts disproving over 200 of Galen's errors, proving the human jawbone is one piece and the breastbone has three segments.

1628 • William Harvey Publishes De Motu Cordis

Key Topic 2.4

Royal physician William Harvey publishes On the Motion of the Heart. Through dissection of cold-blooded reptiles and mathematical calculations, he proves that the heart acts as a mechanical pump recirculating blood continuously through one-way pocket valves.

1660 • Foundation of the Royal Society & Nullius in Verba

Key Topic 2.1

The Royal Society is founded at Gresham College, London, receiving a royal charter from Charles II in 1662. Adopting the motto Nullius in Verba ('Take nobody's word for it'), it launches Philosophical Transactions in 1665 to publish and peer-review scientific experiments.

Vesalius proved Galen had been dissecting pigs, dogs, and apes for 1,300 years. The human jawbone is one piece, not two.

Living Timeline Dual-Coding Protocol: Draw visual symbols for clinical bedside observation, London pest houses, microscope animalcules, and Sydenham's disease classification across the timeline below.

1661 • Marcello Malpighi Observes Capillaries Under the Microscope

Key Topic 2.4

Using an early compound microscope, Italian biologist Marcello Malpighi observes blood passing through microscopic capillaries connecting arteries to veins in a frog's lungs. This provides the definitive optical proof for William Harvey's circulatory model.

1665 • The Great Plague of London: Red Crosses & Pest Houses

Key Topic 2.5

Bubonic plague strikes London, killing approximately 100,000 people (20% of the city). The Lord Mayor orders watchmen to lock infected families inside houses marked with red crosses and 'Lord have mercy upon us'. Over 40,000 stray dogs and cats are slaughtered.

1665 • Robert Hooke Publishes Micrographia

Key Topic 2.1

Robert Hooke presents detailed microscopic drawings of fleas, cork cells, and fly eyes to the Royal Society. While microscopes prove the existence of an invisible world, lenses are too weak to reveal bacteria or germs, leaving medical treatments largely unchanged.

1676 • Thomas Sydenham Publishes Observationes Medicae

Key Topic 2.2

Known as the 'English Hippocrates', Thomas Sydenham urges doctors to base treatments on bedside clinical observation. In Observationes Medicae, he classifies illnesses into distinct species, distinguishes measles from scarlet fever, and popularises cinchona bark (quinine) for malaria.

How did new technologies and scientific societies challenge traditional medical authority?

Core Specification Focus: Medical Renaissance: Humanism; the printing press (c1440); the Royal Society (1660); microscopes and scientific observation.

1 c1440–1476 • Why did printing break the clerical monopoly on medical learning? [1.1]

• Casting lead type in Gutenberg's German workshop	
• Setting up William Caxton's Westminster press	
• Printing hundreds of identical illustrated medical books	

2 c1500 • How did humanism inspire physicians to challenge Galen? [1.2]

• Translating original Greek manuscripts directly	
• Spotting errors made by medieval monk copyists	
• Urging independent questioning of ancient authority	

3 1660 • Why was experimental observation central to the Royal Society? [1.3]

• Meeting at Gresham College in London	
• Receiving royal charter from King Charles II	
• Adopting Latin motto "Nullius in Verba"	

4 1665 • How did scientific journals speed up the exchange of medical knowledge? [1.4]

• Publishing Henry Oldenburg's monthly scientific journal	
• Distributing experimental reports across Europe	
• Debating laboratory findings among rival scientists	

5 1665–1670s • Why did 17th-century microscopes fail to transform medical treatment? [1.5]

• Grinding curved glass lenses in brass cylinders	
• Sketching flea legs in Hooke's Micrographia	
• Observing tiny swimming creatures in pond water	

Gutenberg gave the world movable type; the Royal Society gave the world peer review. Neither cured a cold, but both destroyed ignorance.

Question 4: Explain Why [12 marks • 15 mins]

Explain why new ideas in medicine were able to spread more rapidly in the period c1500–c1700. [12 marks]

Stimulus: **The invention of the printing press (c1440)** | **The Royal Society (1660)** *(You must also use information of your own)*

ACCESSIBLE WRITING FRAMES:

One main reason was that... • For example, ... • This was important because... • Another key factor was... • In particular, ... • Overall, the most decisive reason was... because...

KEY VOCABULARY BANK:

Gutenberg press (c1440) • William Caxton (1476) • clerical monopoly • vernacular translations • Royal Society (1660) • Royal Charter (1662) • Nullius in Verba • Philosophical Transactions (1665) • peer review • animalcules • Robert Hooke (1665)

Write your response in the space provided below:

(Response continues on facing page)

TIMELINE MISSION Turn to Pages 2–3 (Key Topic 2.1). In the drawing box, sketch Caxton's printing press beside the Royal Society's coat of arms (motto: Nullius in Verba). Annotate how printing removed the Church's control over book production. **[ ] Done**

Continue your response to Question 4 below:

• **QUESTION 3: EXPLAIN ONE SIMILARITY [4 MARKS]**

SCORE: [/ 4] • 5 MINS

Explain one way in which ideas about the cause of disease in the Renaissance (c1500–c1700) were similar to ideas in the Medieval period (c1250–c1500). [4 marks]

Focus on persistent beliefs in miasma, supernatural forces, and humoral imbalance despite new anatomical knowledge.

Explain the mechanism: the Royal Society's journal (Philosophical Transactions) allowed European scientists to confirm and repeat experiments.

Why was Thomas Sydenham known as the 'English Hippocrates'?

Core Specification Focus: Thomas Sydenham ('The English Hippocrates'); bedside observation; classification of diseases (Observationes Medicae, 1676); treatments.

1 1660s • Why did Sydenham believe bedside observation surpassed textbook learning? [2.1]

• Visiting sick London patients in private bedrooms	
• Ignoring ancient scholastic university lectures	
• Recording clinical symptoms in detailed notebooks	

2 1676 • How did classifying diseases challenge traditional individual humoral theory? [2.2]

• Classifying distinct illnesses like modern botanists	
• Distinguishing measles from scarlet fever rashes	
• Publishing authoritative textbook used for two centuries	

3 c1670s • Why was the success of Peruvian bark a mortal blow to humoral balance? [2.3]

• Importing bitter Peruvian tree bark to London	
• Dosing marshland ague fevers with powdered bark	
• Proving specific remedies cured specific diseases	

4 c1670s • Why did Sydenham's cool treatment drastically cut smallpox deaths? [2.4]

• Throwing off heavy baronial wool blankets	
• Opening window shutters to let in cold wind	
• Giving cool drinks instead of scalding wines	

5 1680s • Why did medical treatments remain mostly medieval despite Sydenham? [2.5]

• Continuing to believe miasma caused epidemics	
• Lacking laboratory tools to identify bacteria	
• Leaving ordinary patient treatment largely unchanged	

Question 4: Explain Why [12 marks • 15 mins]

EXTENDED WRITING ASSESSMENT

Explain why Thomas Sydenham was an important figure in the development of medicine in the seventeenth century. [12 marks]

Stimulus: **Observationes Medicae (1676)** | **Treatments for smallpox and malaria (cinchona bark)** *(You must also use information of your own)*

ACCESSIBLE WRITING FRAMES:

One main reason was that... • For example, ... • This was important because... • Another key factor was... • In particular, ... • Overall, the most decisive reason was... because...

KEY VOCABULARY BANK:

Thomas Sydenham • English Hippocrates • bedside clinical observation • Observationes Medicae (1676) • species of disease • scarlet fever vs. measles • Cinchona bark (quinine) • cool therapy for smallpox • laudanum

Write your response in the space provided below:

(Response continues on facing page)

TIMELINE MISSION Turn to Pages 2–3 (Key Topic 2.2). In the drawing box, sketch Sydenham taking clinical bedside notes beside a patient with cool fresh air flowing through an open window. Annotate why his classification of diseases was revolutionary. **[ ] Done**

Cool therapy for smallpox: opening windows and throwing off blankets. Radical in 1670, entirely sensible today.

Continue your response to Question 4 below:

• **QUESTION 3: EXPLAIN ONE DIFFERENCE [4 MARKS]** **SCORE: [/ 4] • 5 MINS**

Explain one way in which ideas about the treatment of disease in the Renaissance (c1500–c1700) were different from the Medieval period (c1250–c1500). [4 marks]

Focus on Sydenham's specific treatments (e.g. Peruvian bark for malaria) versus universal humoral purging.

Explain the impact: by proving cinchona bark cured only ague (malaria), Sydenham proved specific remedies could target specific diseases.

How did Andreas Vesalius revolutionize the study of human anatomy?

Core Specification Focus: Andreas Vesalius; The Fabric of the Human Body (De Humani Corporis Fabrica, 1543); human dissection; correcting Galen's errors.

1 1537 • Why was a professor dissecting human bodies revolutionary in 1537? [3.1]

• Climbing down from high lecturer chairs	
• Picking up steel dissection scalpels directly	
• Opening executed criminal bodies on wooden slabs	

2 1540 • How did Vesalius prove Galen had made anatomical mistakes? [3.2]

• Showing human jawbones consist of one piece	
• Proving human breastbones have three segments	
• Demonstrating Galen dissected pigs and Barbary apes	

3 1543 • Why was The Fabrica considered a masterpiece of Renaissance science? [3.3]

• Printing seven massive anatomical volumes in Basel	
• Engraving precise woodcut plates of human muscles	
• Dedication presented to Holy Roman Emperor Charles V	

4 1545–1553 • How did English translations spread Vesalius's ideas to barber-surgeons? [3.4]

• Copying Vesalius's woodcuts onto English copper plates	
• Translating Latin descriptions into everyday English	
• Selling anatomy manuals to London barber-surgeons	

5 c1560 • Why did Vesalius's anatomical discoveries fail to save sick patients immediately? [3.5]

• Facing angry attacks from conservative university professors	
• Resigning academic post to serve royal courts	
• Acknowledging anatomy did not yet cure living sickness	

Vesalius proved Galen had been dissecting pigs, dogs, and apes for 1,300 years. The human jawbone is one piece, not two. Awkward for Galenists!

Question 4: Explain Why [12 marks • 15 mins]

Explain why Andreas Vesalius made significant breakthroughs in the understanding of human anatomy in the sixteenth century. [12 marks]

Stimulus: **De Humani Corporis Fabrica (1543)** | **Dissection of human corpses rather than animals** *(You must also use information of your own)*

ACCESSIBLE WRITING FRAMES:

One main reason was that... • For example, ... • This was important because... • Another key factor was... • In particular, ... • Overall, the most decisive reason was... because...

KEY VOCABULARY BANK:

Andreas Vesalius • Padua University • human dissection • De Humani Corporis Fabrica (1543) • anatomical woodcuts • jawbone (mandible) • breastbone segments • Barbary apes vs. humans • Thomas Geminus

Write your response in the space provided below:

(Response continues on facing page)

TIMELINE MISSION Turn to Pages 2–3 (Key Topic 2.3). In the drawing box, sketch Vesalius standing over an anatomical dissection table holding a scalpel. Annotate two specific errors of Galen that he disproved. **[ ] Done**

Continue your response to Question 4 below:

• **QUESTION 3: EXPLAIN ONE SIMILARITY [4 MARKS]** **SCORE: [/ 4] • 5 MINS**

Explain one way in which methods of anatomical training in the Renaissance (c1500–c1700) were similar to anatomical training in the nineteenth century. [4 marks]

Focus on direct physical dissection of human bodies as the foundation of surgical education.

Explain the limitation: while Vesalius proved Galen wrong on over 200 anatomical points, his discoveries did not immediately cure a single disease.

How did William Harvey prove the circulation of the blood, and why was it resisted?

Core Specification Focus: William Harvey; circulation of the blood; De Motu Cordis (1628); the heart as a mechanical pump; veins and one-way valves; limitations.

1 c1615 • Why did Harvey study cold-blooded animals to understand the human heart? [4.1]

• Observing slow heartbeats in live cold frogs	
• Dissecting human corpses at the Royal College	
• Mapping one-way pocket valves inside veins	

2 1620s • How did mathematical calculations prove Galen's theory impossible? [4.2]

• Comparing the beating heart to mechanical water pumps	
• Calculating liver could not produce gallons of blood daily	
• Proving the same blood circulates continuously	

3 1628 • Why did Harvey publish his research abroad rather than in England? [4.3]

• Publishing Latin treatise on heart motion in Frankfurt	
• Demonstrating tied forearm vein experiments	
• Dedicating medical breakthrough to King Charles I	

4 1630s–1650s • Why was Harvey's discovery fiercely rejected by established doctors? [4.4]

• Being mocked as a madman by rival doctors	
• Losing wealthy private London medical patients	
• Waiting fifty years for Oxford university to teach it	

5 1661 • Why was the discovery of capillaries the final proof Harvey was right? [4.5]

• Focusing microscope lenses on live frog lungs	
• Discovering microscopic capillary vessels connecting arteries	
• Confirming Harvey's circulatory model beyond all doubt	

Galen thought the liver made fresh blood like a factory; Harvey calculated you would need to produce 250 litres a day. Mathematics: the ultimate Galen-slayer.

Question 4: Explain Why [12 marks • 15 mins]

EXTENDED WRITING ASSESSMENT

Explain why William Harvey's discovery of the circulation of the blood was a major breakthrough in medicine in the seventeenth century. [12 marks]

Stimulus: **De Motu Cordis (1628)** | **Mechanical pump calculations and vein experiments** *(You must also use information of your own)*

ACCESSIBLE WRITING FRAMES:

One main reason was that... • For example, ... • This was important because... • Another key factor was... • In particular, ... • Overall, the most decisive reason was... because...

KEY VOCABULARY BANK:

William Harvey • De Motu Cordis (1628) • circulation of the blood • mechanical pump • one-way pocket valves • tied-arm ligature experiment • quantitative calculation • Marcello Malpighi (1661) • capillaries

Write your response in the space provided below:

(Response continues on facing page)

TIMELINE MISSION Turn to Pages 2–3 (Key Topic 2.4). In the drawing box, sketch Harvey's tied-arm ligature experiment showing blood pooling at the valves. Annotate his mathematical calculation proving Galen's liver-blood theory was impossible. **[ ] Done**

Harvey tied a bandage around an arm and watched the veins swell. Simple experiment, monumental consequence for human physiology.

Continue your response to Question 4 below:

• **QUESTION 3: EXPLAIN ONE DIFFERENCE [4 MARKS]** **SCORE: [/ 4] • 5 MINS**

Explain one way in which ideas about the human body in the Renaissance (c1500–c1700) were different from the Medieval period (c1250–c1500). [4 marks]

Contrast Harvey's dynamic mechanical circulation model with Galen's belief that the liver manufactured fresh blood consumed by organs.

Why did treatments during the Great Plague of 1665 show so much continuity with 1348?

Core Specification Focus: Continuity in treatment and prevention; the Great Plague of London (1665); comparison with the Black Death; plague orders; quarantine.

1 Spring 1665 • Why did the rich flee London while the poor suffered catastrophic mortality? [5.1]

- **Spreading through poor crowded London parishes**
- **Fleeing town by carriage: King Charles II and physicians**
- **Leaving poor weavers trapped behind city gates**

2 Summer 1665 • Why did people still use 14th-century remedies in 1665? [5.2]

- **Blaming planetary alignments and wrath of God**
- **Sniffing dried sponges soaked in vinegar**
- **Strapping plucked live pigeons to hot buboes**

3 1665 • How did local government action in 1665 show greater organization than 1348? [5.3]

- **Painting red crosses on barred front doors**
- **Posting armed watchmen to enforce quarantine**
- **Building wooden pest houses outside city walls**

4 1665 • Why was slaughtering cats and dogs a tragic medical mistake? [5.4]

- **Killing forty thousand domestic dogs and cats**
- **Burning barrels of black tar at street crossings**
- **Smoking tobacco pipes to ward off foul miasma**

5 Winter 1665 • How devastating was the demographic impact of the 1665 epidemic? [5.5]

- **Ringing bell carts through silent dark lanes**
- **Calling "Bring out your dead" into empty alleys**
- **Burying one hundred thousand Londoners in mass trenches**

1665: They killed 40,000 dogs and cats to stop the plague. The real culprits—the rat fleas—enjoyed a predator-free paradise.

Question 5/6: Thematic Essay [16 marks + 4 marks SPaG • 25 mins]

EXTENDED WRITING ASSESSMENT

'Approaches to dealing with the Great Plague in 1665 were completely different from approaches to the Black Death in 1348.' How far do you agree? Explain your answer. [16 marks + 4 marks SPaG]

Stimulus: **Mayor's plague orders and quarantine (red crosses)** | **Beliefs in miasma, astrology, and supernatural causes** *(You must also use information of your own)*

ACCESSIBLE WRITING FRAMES:

One main reason was that... • For example, ... • This was important because... • Another key factor was... • In particular, ... • Overall, the most decisive reason was... because...

KEY VOCABULARY BANK:

Great Plague of London (1665) • Lord Mayor's orders • searchers of the dead • red crosses • "Lord have mercy upon us" • quarantine • pest houses • slaughter of 40,000 dogs and cats • smoking tar barrels • dead carts

Write your response in the space provided below:

(Response continues on facing page)

TIMELINE MISSION Turn to Pages 2–3 (Key Topic 2.5). In the drawing box, sketch a London front door marked with a red cross and guarded by a watchman. Annotate two continuities and two differences between 1348 and 1665. **[ ] Done**

Red crosses on doors, armed watchmen, and smoking tar barrels. London in 1665 was a nightmare of disease, quarantine, and municipal panic.

Continue your response to Question 5/6 below:

• **QUESTION 3: EXPLAIN ONE DIFFERENCE [4 MARKS]** SCORE: [/ 4] • 5 MINS

Explain one way in which government responses to the Great Plague of London (1665) were different from government responses to the Black Death (1348). [4 marks]

Contrast systematic quarantine, red crosses, and pest houses in 1665 with rudimentary 1349 street cleaning.

KEY TOPIC 2 NOTES & EXAMINATION STRATEGY

Notes: Record key causal links, essay plans, and teacher feedback for this unit.

PUPIL NAME: _____ TARGET: _____ CURRENT GRADE: _____

ENQUIRY	ASSESSMENT COMPONENT FOCUS	PAGE	SCORE	TEACHER COMMENT & NEXT STEPS
KT2.1	**Q4:** Explain Why New Medical Ideas Spread More Rapidly c1500–c1700	p. 5–6	/ 12	
	Q3: Explain One Similarity (Ideas on Cause: Renaissance vs Medieval)	p. 6	/ 4	
KT2.2	**Q4:** Explain Why Thomas Sydenham Was an Important Figure	p. 8–9	/ 12	
	Q3: Explain One Difference (Treatments: Renaissance vs Medieval)	p. 9	/ 4	
KT2.3	**Q4:** Explain Why Andreas Vesalius Made Anatomical Breakthroughs	p. 11–12	/ 12	
	Q3: Explain One Similarity (Anatomical Training: Renaissance vs 19th C)	p. 12	/ 4	
KT2.4	**Q4:** Explain Why William Harvey's Circulation Discovery Was Major Breakthrough	p. 14–15	/ 12	
	Q3: Explain One Difference (Ideas About Human Body: Renaissance vs Medieval)	p. 15	/ 4	
KT2.5	**Q5/6:** Evaluative Essay: Great Plague (1665) vs Black Death (1348)	p. 17–18	/ 20	
	Q3: Explain One Difference (Government Responses: 1665 vs 1348)	p. 18	/ 4	
TOTAL	**Key Topic 2 Cumulative Examination Assessment Portfolio (10 Tasks)**	p. 4–18	/ 88	**Overall Grade:** _____ **Teacher Target:** _____

• DIGITAL LEARNING & RETRIEVAL HUB • INTERACTIVE LESSONS & QUIZZES

Interactive Textbook • 10 Questions Per Enquiry

KT2.1	KT2.2	KT2.3	KT2.4	KT2.5
New Spirit	Thomas Sydenham	Andreas Vesalius	William Harvey	Great Plague
LESSON HUB	LESSON HUB	LESSON HUB	LESSON HUB	LESSON HUB
PRACTICE QUIZ	PRACTICE QUIZ	PRACTICE QUIZ	PRACTICE QUIZ	PRACTICE QUIZ
Score: [/ 10]	Score: [/ 10]	Score: [/ 10]	Score: [/ 10]	Score: [/ 10]

Nullius in Verba: "Take nobody's word for it." Scanning these quiz QR codes will prove whether your memory holds up under exam pressure.

PUPIL NAME: _____ TARGET: _____ CLASS: _____

KEY TOPIC 3 • C1700–C1900 CHRONOLOGICAL ENQUIRY SEQUENCE

Medicine in 18th & 19th Century Britain

c1700–c1900: Ideas on Causes, Prevention, Hospital Care, Surgery & Public Health



HISTORICAL PRIMARY RECORD • 1802 **HISTORICAL SATIRE ARCHIVE**

James Gillray, The Cow-Pock—or—the Wonderful Effects of the New Inoculation! (1802)

Hand-coloured etching lampooning early Victorian fears that Edward Jenner's cowpox vaccine would cause patients to sprout bovine horns and calves.

PRIMARY VISUAL EVIDENCE **THE BRITISH MUSEUM DEPARTMENT OF PRINTS & DRAWINGS**

KEY TOPIC 3 SPECIFICATION FRAMEWORK (EDEXCEL 1HI0/11)	LESSON DONE	QUIZ [10]
KT3.1: Ideas on Causes: Pasteur's Germ Theory & Koch's Microbe Hunters (c.1861–c.1883) *Why did it take over two decades for British doctors to accept Louis Pasteur's Germ Theory?* **SPEC:** Continuity and change in explanations of the cause of disease and illness. The influence in Britain of Pasteur's Germ Theory and Koch's work on microbes	☐	☐
KT3.2: Approaches to Prevention: Edward Jenner & The Smallpox Vaccine (1796–1853) *Why did Edward Jenner's smallpox vaccine face ferocious opposition despite proving scientifically successful?* **SPEC:** Key individual: Jenner and the development of vaccination. New approaches to prevention: the development and use of vaccinations	☐	☐
KT3.3: Improvements in Hospital Care: Florence Nightingale, Mary Seacole & The Sanitary Revolution (1854–1863) *How did Florence Nightingale transform British hospitals from Victorian death houses into clean, scientific institutions?* **SPEC:** The extent of change in care and treatment: improvements in hospital care and the influence of Nightingale on nursing and hospitals in Britain	☐	☐
KT3.4: The Surgical Revolution: Simpson, Lister & The Conquest of Pain and Sepsis (1847–1870s) *Why did the discovery of anaesthetics initially make surgical operations more dangerous rather than safer?* **SPEC:** The impact of anaesthetics and antiseptics on surgery: Simpson's chloroform, Lister's carbolic acid, and the development of aseptic surgery	☐	☐
KT3.5: Public Health & Cholera: John Snow, The Broad Street Pump & The 1875 Public Health Act (1854–1875) *Why did the British government abandon its laissez-faire policy to pass the compulsory 1875 Public Health Act?* **SPEC:** Fighting Cholera in London (1854); attempts to prevent its spread; the significance of Snow and the Broad Street pump. The Public Health Act (1875)	☐	☐

Living Timeline • Part 1: Vaccines, Anaesthetics & Early Sanitation (1700–1854)

Living Timeline Dual-Coding Protocol: Complete each milestone sketchpad using dual-coding (combining visual storytelling symbols, causal arrows, and dates) to master the chronological spine across the centre-fold.

1796 • Edward Jenner: Smallpox Vaccination & Immunology

Key Topic 3.2

Jenner inoculates James Phipps with cowpox pus from Sarah Nelmes, proving cowpox confers immunity against smallpox. Publishes findings in 1798. Despite initial Royal Society scepticism, vaccination eradicates inoculator infection risks and founds immunology.

1847 • James Young Simpson: Chloroform & Anaesthesia

Key Topic 3.4

Simpson discovers the potent anaesthetic properties of chloroform in Edinburgh, ending agony during surgery. Queen Victoria's use of chloroform during childbirth (1853) popularises it, enabling complex operations despite the initial 'Black Period' of surgery.

1848 • First Public Health Act & The General Board of Health

Key Topic 3.5

Spurred by Chadwick's 1842 report and epidemic cholera, Parliament passes the permissive 1848 Act. Towns can create local boards of health to supply clean water and drainage, though lack of compulsion leaves most cities vulnerable to filth and contagion.

1854 • John Snow: Broad Street Pump & Waterborne Cholera

Key Topic 3.5

During the Soho epidemic, Snow compiles his famous spot map linking 500+ fatal cholera cases to the Broad Street pump. Removing the handle immediately halts the outbreak, mathematically disproving the miasma theory seven years before Pasteur's Germ Theory.

The 18th century relied on empirical observation; the 19th century unlocked the laboratory.

Living Timeline • Part 2: Nursing, Microbes & Antiseptics (1854–1875)

CHRONOLOGICAL INQUIRY • MILESTONES 5–8

Living Timeline Dual-Coding Protocol: Draw visual symbols for Crimean hospital sanitation, swan-neck flasks, Lister carbolic sprays, and municipal sewers across the timeline below.

1854–56 • Florence Nightingale: Hospital Sanitation at Scutari

Key Topic 3.3

During the Crimean War, Nightingale enforces rigorous cleanliness, ventilation, and fresh nutrition at Scutari hospital, slashing soldier death rates from 40% to 2%. Returns to Britain to establish the Nightingale Training School and author Notes on Nursing (1859).

1861 • Louis Pasteur: Germ Theory & Swan-Neck Flask Experiments

Key Topic 3.1

Commissioned by Lille brewers, Pasteur disproves spontaneous generation using swan-neck flasks, demonstrating that airborne microbes cause fermentation and decay. In 1861, he publishes Germ Theory, proving microscopic germs cause specific diseases in living organisms.

1865–67 • Joseph Lister: Carbolic Acid Spray & Antiseptic Surgery

Key Topic 3.4

Applying Pasteur's Germ Theory to wound sepsis, Lister uses carbolic acid sprays, dressings, and hand-washing during surgery at Glasgow Infirmary. His antiseptic techniques reduce amputee mortality from 46% to 15%, paving the way for modern aseptic operating theatres.

1875 • The Second Public Health Act: End of Laissez-Faire

Key Topic 3.5

Disraeli's government ends laissez-faire by making public health compulsory. Municipal councils are legally required to provide clean piped water, subterranean sewer drainage, street lighting, and medical officers of health, transforming urban life expectancy.

Why did it take 30 years for Germ Theory to transform medical understanding in Britain?

Core Specification Focus: Germ Theory (1861); Louis Pasteur; Robert Koch; bacteriology; agar cultures, chemical dyes, and microscopes; impact on Britain.

1 c1700–1850s • Why did doctors believe disease created germs, rather than vice versa? [1.1]

• Smelling rotting meat and open cesspools	
• Foul airborne damp blamed for illness	
• Microscopes seeing tiny swimming creatures	

2 1861 • How did the swan-neck flask disprove spontaneous generation? [1.2]

• Heating sour beer inside glass flasks	
• Bent swan necks trapping airborne dust	
• Boiled broth staying clear and fresh	

3 1867–1876 • Why did British doctors refuse to believe germs caused human illness? [1.3]

• London doctors defending bad air theory	
• Shining light beams through dusty boxes	
• Spraying smelly carbolic acid on cut flesh	

4 1876–1882 • How did Koch's staining and culture methods prove specificity? [1.4]

• Solid seaweed agar jelly in shallow dishes	
• Staining microbes with bright purple dyes	
• Photographing deadly tuberculosis under glass lenses	

5 1883–1890 • Why was photomicrography crucial in winning British doctors over? [1.5]

• Tracking curved cholera germs in river water	
• Taking crisp black-and-white photos of germs	
• Injecting farm animals to test poisons	

Pasteur proved that rotting soup doesn't spontaneously create germs; Koch proved which specific germs caused tuberculosis.

Question 4: Explain Why [12 marks • 15 mins]

EXTENDED WRITING ASSESSMENT

Explain why there was rapid progress in understanding the causes of disease in the period c1860–c1890. [12 marks]

Stimulus: **Louis Pasteur's Germ Theory (1861)** **Robert Koch's work on bacteria** *(You must also use information of your own)*

ACCESSIBLE WRITING FRAMES:

One main reason was that... • For example, ... • This was important because... • Another key factor was... • In particular, ... • Overall, the most decisive reason was... because...

KEY VOCABULARY BANK:

swan-neck flask • Germ Theory (1861) • spontaneous generation • Robert Koch • agar jelly • methyl violet dye • tuberculosis (1882) • microphotography

Write your response in the space provided below:

(Response continues on facing page)

TIMELINE MISSION Turn to Pages 2–3 (Key Topic 3.1). In the drawing box, sketch Pasteur's swan-neck flask experiment and Koch's agar plate. Annotate how identifying specific bacteria ended miasma theory forever! **[ ] Done**

Continue your response to Question 4 below:

• **QUESTION 3: EXPLAIN ONE DIFFERENCE [4 MARKS]** **SCORE: [/ 4] • 5 MINS**

Explain one way in which ideas about the cause of disease in the late nineteenth century (c1860–c1890) were different from ideas in the Renaissance (c1500–c1700). [4 marks]

Contrast laboratory discoveries of specific microbes with earlier beliefs in miasma and spontaneous generation.

Why did Edward Jenner face fierce resistance despite discovering the smallpox vaccine?

Core Specification Focus: Edward Jenner; development of smallpox vaccination (1796); opposition from doctors, church, and variolators; government funding and compulsory vaccination (1852).

1 1721–1795 • Why was inoculation dangerous for poor urban communities? [2.1]

- **Scabbing sores blinding young children**
- **Scratching live pus into healthy arms**
- **Paying rich doctors high guinea fees**

2 1796–1798 • How did Jenner use scientific experimentation to test folklore? [2.2]

- **Milking cows with clear smooth skin**
- **Scraping blister liquid into boy James Phipps**
- **Dosing boy with deadly pox without illness**

3 1798–1810s • Why did London doctors and church leaders resist vaccination? [2.3]

- **Printing scary cartoons with sprouting cow horns**
- **Church pulpits warning against beast blood**
- **Dirty hospital needles passing deadly infections**

4 1802–1840 • Why did the British government abandon laissez-faire for vaccination? [2.4]

- **Voting thirty thousand pounds in gold coins**
- **Giving free scratches to village babies**
- **Banning old dangerous pus cuts by law**

5 1852–1871 • Why did making vaccination compulsory spark fierce public protests? [2.5]

- **Fining poor parents for refusing jabs**
- **Angry street protests marching with banners**
- **Plunging death rates as smallpox disappears**

Cartoons showed vaccinated people growing cow heads and horns: proof that anti-vax memes existed long before social media.

Question 5/6: Evaluative Essay [16+4 marks • 20 mins]

EXTENDED WRITING ASSESSMENT

'Edward Jenner's development of the smallpox vaccine was the most significant breakthrough in the prevention of disease in the period c1700–c1900.' How far do you agree? [16+4 marks]

Stimulus: **Inoculation** **The Public Health Act (1875)** *(You must also use information of your own)*

ACCESSIBLE WRITING FRAMES:

On the one hand, a major breakthrough was... • For example, ... • This was significant because... • On the other hand, an alternative factor was... • In particular, ... • Overall, the most decisive factor was... because...

KEY VOCABULARY BANK:

Edward Jenner (1796) • cowpox • James Phipps • smallpox • variolation • inoculation • compulsory vaccination (1852) • Anti-Vaccination League

Write your response in the space provided below:

(Response continues on facing page)

TIMELINE MISSION Turn to Pages 2–3 (Key Topic 3.2). In the drawing box, sketch Jenner inoculating James Phipps with cowpox. Annotate how this empirical experiment laid the foundation for modern immunology! **[ ] Done**

Jenner saved millions with cowpox, yet died before anyone could explain what a virus actually was.

Continue your response to Question 5/6 below:

• **QUESTION 3: EXPLAIN ONE DIFFERENCE [4 MARKS]** **SCORE: [/ 4] • 5 MINS**

Explain one way in which methods of preventing smallpox in the late eighteenth century were different from methods used in the seventeenth century. [4 marks]

Contrast safe cowpox vaccination with earlier reliance on prayer, quarantine, and risky variolation.

How did Florence Nightingale transform hospitals from death traps into places of cure?

Core Specification Focus: Florence Nightingale; hospital conditions at Scutari (1854–56); Notes on Hospitals and Notes on Nursing (1859); pavilion hospital plan; professional nursing training.

1 c1700–1850 • Why were early Victorian hospital wards notorious death traps? [3.1]

- **Crowded dark rooms smelling of festering wounds**
- **Drunk ward women sleeping on dirty rags**
- **Sharing blood-soaked blankets between beds**

2 1854–1856 • How did Nightingale's sanitary reforms drastically cut the death rate? [3.2]

- **Sewers overflowing beneath wooden ward floors**
- **Scrubbing dirty walls with boiling water and soap**
- **Feeding wounded soldiers hot fresh stew**

3 1859 • Why did Nightingale's belief in miasma still produce effective hygiene? [3.3]

- **Opening high windows for fresh river breezes**
- **Drawing colourful polar pie charts for MPs**
- **Demanding clean white cotton bandages**

4 1860s • How did the pavilion plan prevent cross-infection between wards? [3.4]

- **Building long narrow brick ward pavilions**
- **Washing smooth glazed walls with water**
- **Cross-ventilation blowing stale room air away**

5 1860–1890s • How did nurse training permanently raise the standard of hospital care? [3.5]

- **Uniformed women learning medical rules at St Thomas'**
- **Stern matrons inspecting spotless bedside tables**
- **Fusing clean hands with microbe science**

Nightingale believed miasma caused disease, yet her obsession with fresh air, scrubbing, and clean drains saved thousands of lives.

Question 4: Explain Why [12 marks • 15 mins]

EXTENDED WRITING ASSESSMENT

Explain why the standard of hospital care and nursing improved so significantly in the second half of the nineteenth century. [12 marks]

Stimulus: **Florence Nightingale's work in the Crimea (1854–56)** | **The pavilion hospital plan** *(You must also use information of your own)*

ACCESSIBLE WRITING FRAMES:

One main reason was that... • For example, ... • This was important because... • Another key factor was... • In particular, ... • Overall, the most decisive reason was... because...

KEY VOCABULARY BANK:

Florence Nightingale • Scutari (1854) • Notes on Nursing (1859) • pavilion plan • cross-ventilation • St Thomas' Hospital • trained nurses • sanitation

Write your response in the space provided below:

(Response continues on facing page)

TIMELINE MISSION Turn to Pages 2–3 (Key Topic 3.3). In the drawing box, sketch the Pavilion Hospital plan with its separate, ventilated pavilions. Annotate how cross-ventilation prevented hospital gangrene! **[ ] Done**

Continue your response to Question 4 below:

• **QUESTION 3: EXPLAIN ONE DIFFERENCE [4 MARKS]** **SCORE: [/ 4] • 5 MINS**

Explain one way in which hospital care in the late nineteenth century was different from hospital care in the Medieval period. [4 marks]

Contrast modern sanitary pavilion wards and trained medical care with medieval monastic care.

Why did the conquest of pain in surgery temporarily increase the patient death rate?

Core Specification Focus: Surgical breakthroughs: pain (anaesthetics: nitrous oxide, ether, James Simpson and chloroform); infection (Joseph Lister and carbolic acid; asepsis); bleeding.

1 c1700–1846 • Why was surgical speed essential before the discovery of anaesthetics? [4.1]

- **Holding down screaming patients with leather straps**
- **Sawing off broken legs in thirty seconds**
- **Breathing sweet laughing gas at fairgrounds**

2 1847–1853 • Why did the death of Hannah Greener cause panic among surgeons? [4.2]

- **Sniffing heavy liquid vapour under dinner table**
- **Young Hannah Greener dying from cold overdose**
- **Queen Victoria giving birth with zero pain**

3 1846–1865 • Why did conquering surgical pain lead to higher patient death rates? [4.3]

- **Wearing stiff dried blood coats as badges**
- **Cutting deeper into belly flesh for longer**
- **Black rotting gangrene killing feverish patients**

4 1865–1870s • How did Lister apply Germ Theory to solve hospital gangrene? [4.4]

- **Treating green sewage water with smelly chemicals**
- **Wrapping boy Jamie Greenlees' fractured leg**
- **Pumping fine acid mist over open surgical wounds**

5 1880s–1890s • What is the crucial difference between antiseptic and aseptic surgery? [4.5]

- **Boiling metal scalpels in pressurized steam kettles**
- **Pulling on thin sterile rubber gloves**
- **Tying white cotton masks over mouths and noses**

Question 4: Explain Why [12 marks • 15 mins]

EXTENDED WRITING ASSESSMENT

Explain why surgery became significantly safer in the period c1840–c1900. [12 marks]

Stimulus: **James Simpson's discovery of chloroform (1847)** | **Joseph Lister's use of carbolic acid (1865)** *(You must also use information of your own)*

ACCESSIBLE WRITING FRAMES:

One main reason was that... • For example, ... • This was important because... • Another key factor was... • In particular, ... • Overall, the most decisive reason was... because...

KEY VOCABULARY BANK:

James Simpson • chloroform (1847) • Black Period • Joseph Lister • carbolic acid spray (1865) • antiseptics • steam sterilisation • aseptic surgery

Write your response in the space provided below:

(Response continues on facing page)

TIMELINE MISSION Turn to Pages 2–3 (Key Topic 3.4). In the drawing box, sketch Lister's carbolic donkey engine spraying mist over an open wound. Annotate the difference between antiseptic and aseptic surgery! **[ ] Done**

Surgeons used to take pride in blood-caked frock coats; Lister forced them to wash their hands and operate in carbolic acid mist.

Continue your response to Question 4 below:

• **QUESTION 3: EXPLAIN ONE SIMILARITY [4 MARKS]** SCORE: [/ 4] • 5 MINS

Explain one way in which surgical operations in the early nineteenth century (before 1846) were similar to surgery in the Medieval period. [4 marks]

Focus on raw surgical speed and physical restraint before effective chemical anaesthetics were discovered.

Why did it take 20 years for John Snow's cholera discovery to change government policy?

Core Specification Focus: Public health: cholera epidemics; John Snow and Broad Street pump (1854); public health acts (1848 permissive vs 1875 compulsory); Bazalgette and London sewers.

1 1831–1848 • Why did towns refuse to set up health boards under the 1848 Act? [5.1]

- **Violent watery sickness turning human skin blue**
- **Piles of human excrement rotting near houses**
- **Parliament passing optional town health boards**

2 1854 • How did Snow use epidemiological mapping to prove water transmission? [5.2]

- **Plotting black death bars on street map**
- **Brewery men drinking ale staying completely healthy**
- **Unscrewing heavy iron pump handle with wrench**

3 1854–1858 • Why did the medical establishment refuse to accept Snow's water theory? [5.3]

- **Cracked brick cesspool leaking into drinking well**
- **Medical journals laughing at dirty water theory**
- **Doctors burning pitch barrels in foggy streets**

4 1858–1865 • Why did fear of miasma paradoxically save thousands from cholera? [5.4]

- **Boiling brown river mud halting Parliament debates**
- **Soaking courtroom window curtains in lime juice**
- **Digging thousand miles of underground brick tunnels**

5 1875 • Why was the 1875 Act the definitive death blow to laissez-faire? [5.5]

- **Working factory men winning parliamentary votes**
- **Forcing town councils to pipe clean fresh water**
- **Hiring official medical inspectors for bad food**

Snow removed the Broad Street pump handle in 1854, proving cholera was water-borne seven years before Germ Theory.

Question 5/6: Evaluative Essay [16+4 marks • 20 mins]

EXTENDED WRITING ASSESSMENT

'The work of John Snow was the main reason for improvements in public health in the nineteenth century.' How far do you agree? [16+4 marks]

Stimulus: Edwin Chadwick's 1842 Report | The Public Health Act (1875) *(You must also use information of your own)*

ACCESSIBLE WRITING FRAMES:

On the one hand, a major breakthrough was... • For example, ... • This was significant because... • On the other hand, an alternative factor was... • In particular, ... • Overall, the most decisive factor was... because...

KEY VOCABULARY BANK:

John Snow (1854) • Broad Street pump • waterborne cholera • Edwin Chadwick (1842) • Great Stink (1858) • Joseph Bazalgette • Public Health Act (1875) • laissez-faire

Write your response in the space provided below:

(Response continues on facing page)

TIMELINE MISSION Turn to Pages 2–3 (Key Topic 3.5). In the drawing box, sketch John Snow's Broad Street pump map and Bazalgette's sewer network. Annotate how cholera forced Parliament to abandon laissez-faire! [ ] Done

Continue your response to Question 5/6 below:

• **QUESTION 3: EXPLAIN ONE DIFFERENCE [4 MARKS]** **SCORE: [/ 4] • 5 MINS**

Explain one way in which government action on public health in the late nineteenth century was different from government action in the Renaissance. [4 marks]

Contrast compulsory public health legislation with earlier temporary emergency measures.

KEY TOPIC 3 NOTES & EXAMINATION STRATEGY

Notes: Record key causal links, essay plans, and teacher feedback for this unit.

PUPIL NAME: _____ TARGET: _____ CURRENT GRADE: _____

ENQUIRY	ASSESSMENT COMPONENT FOCUS	PAGE	SCORE	TEACHER COMMENT & NEXT STEPS
KT3.1	**Q4:** Explain Why Rapid Progress in Causes c1860–c1890	p. 5–6	/ 12	
	Q3: Explain One Difference (Causes: c1700 vs c1900)	p. 6	/ 4	
KT3.2	**Q5/6:** Evaluative Essay on Jenner Smallpox Vaccine c1700–c1900	p. 8–9	/ 20	
	Q3: Explain One Difference (Smallpox Prevention Methods)	p. 9	/ 4	
KT3.3	**Q4:** Explain Why Hospital Care & Nursing Improved c1850–c1900	p. 11–12	/ 12	
	Q3: Explain One Difference (Hospital Care: Medieval vs c1900)	p. 12	/ 4	
KT3.4	**Q4:** Explain Why Surgery Became Safer c1840–c1900	p. 14–15	/ 12	
	Q3: Explain One Similarity in Surgery (Medieval vs Pre-1846)	p. 15	/ 4	
KT3.5	**Q5/6:** Evaluative Essay on John Snow & 1875 Public Health Act	p. 17–18	/ 20	
	Q3: Explain One Difference in Public Health Action (1848 vs 1875)	p. 18	/ 4	
TOTAL	**Key Topic 3 Cumulative Examination Assessment Portfolio (10 Tasks)**	p. 4–18	/ 88	**Overall Grade:** _____ **Teacher Target:** _____

• DIGITAL LEARNING & RETRIEVAL HUB • INTERACTIVE LESSONS & QUIZZES

Interactive Textbook • 10 Questions Per Enquiry

KT3.1	KT3.2	KT3.3	KT3.4	KT3.5
Pasteur & Koch	Edward Jenner	F. Nightingale	Simpson & Lister	Snow & 1875 Act
LESSON HUB	LESSON HUB	LESSON HUB	LESSON HUB	LESSON HUB
PRACTICE QUIZ	PRACTICE QUIZ	PRACTICE QUIZ	PRACTICE QUIZ	PRACTICE QUIZ
Score: [/ 10]	Score: [/ 10]	Score: [/ 10]	Score: [/ 10]	Score: [/ 10]

Sanitation, antiseptics, and vaccinations conquered disease; retrieval practice will conquer your GCSE exam.

PUPIL NAME: _____ TARGET: _____ CLASS: _____

KEY TOPIC 4 • C1900–PRESENT CHRONOLOGICAL ENQUIRY SEQUENCE

Medicine in Modern Britain, c1900–present

c1900–present: Ideas on Causes, Diagnostic Technology, The NHS, Penicillin & Lung Cancer



HISTORICAL PRIMARY RECORD • 5 JULY 1948 **COI-INF-12-88**

Aneurin Bevan with First NHS Patient Sylvia Beckingham, Park Hospital

Central Office of Information • Minister of Health Aneurin Bevan inaugurating the National Health Service at Park Hospital, Davyhulme, marking universal free access to healthcare.

PRIMARY VISUAL EVIDENCE **THE NATIONAL ARCHIVES • INF 12/88**

KEY TOPIC 4 SPECIFICATION FRAMEWORK (EDEXCEL 1HI0/11)	LESSON DONE	QUIZ [10]
KT4.1: Ideas on Causes: Genetics, DNA & The Human Genome Project (c.1953–present) *How did unlocking the double helix and mapping the human genome revolutionize our understanding of disease causation?* **SPEC:** Advances in understanding the causes of illness and disease: the influence of genetic and lifestyle factors on health. Improvements in diagnosis: blood tests, scans and monitors	☐	☐
KT4.2: Lifestyle Factors & The Technological Revolution in Diagnosis (c.1900–present) *How did the shift toward lifestyle diseases and the rise of diagnostic scanning transform 20th-century medicine?* **SPEC:** The extent of change in care and treatment: the impact of the NHS and science and technology; improved access to care; magic bullets and antibiotics; high-tech hospital care	☐	☐
KT4.3: The Search for Magic Bullets, High-Tech Treatments & The Birth of the NHS (1909–1948) *How did synthetic magic bullets and the foundation of the NHS dismantle medical inequality in Britain?* **SPEC:** New approaches to prevention: mass vaccinations and government lifestyle campaigns	☐	☐
KT4.4: Case Study 1: The Antibiotic Revolution: Fleming, Florey & Chain and Penicillin (1928–1945) *How did an accidental laboratory mold transform into humanity's first mass-produced antibiotic wonder drug?* **SPEC:** Key individuals: Fleming, Florey and Chain's development of penicillin	☐	☐
KT4.5: Case Study 2: Public Health & The Fight Against Lung Cancer (c.1950–present) *How did the battle against lung cancer drive modern high-tech oncology and permanently dismantle government laissez-faire?* **SPEC:** The fight against lung cancer in the twenty-first century: the use of science and technology in diagnosis and treatment; government action	☐	☐

Living Timeline • Part 1: Magic Bullets, Blood Groups & Early Antibiotics (1900–1940)

Living Timeline Dual-Coding Protocol: Draw visual symbols for ABO agglutination, Salvarsan magic bullets, Fleming's mould petri dish, and Prontosil sulfa dye across the timeline below.

1901 • Karl Landsteiner: ABO Blood Groups Discovered

Key Topic 4.1

Landsteiner identifies A, B, and O blood groups, proving agglutination occurs when incompatible blood types mix. This discovery transforms blood transfusions from fatal gambles into safe, predictable surgical lifelines, foundational for modern trauma surgery.

1909 • Paul Ehrlich & Sahachiro Hata: Salvarsan 606

Key Topic 4.3

Building on dye research, Ehrlich and Hata systematically test hundreds of arsenic compounds, discovering arsphenamine (compound 606). Humanity's first synthetic chemical 'magic bullet' targets syphilis bacteria without poisoning human tissue.

1928 • Alexander Fleming: Accidental Discovery of Penicillin

Key Topic 4.4

Returning from holiday, Fleming notices a contaminated staphylococcus culture dish with a halo of destroyed bacteria around Penicillium notatum mould. Fleming publishes in 1929, though unable to purify or extract enough active substance for clinical trials.

1932 • Gerhard Domagk: Prontosil & Sulfonamide Drugs

Key Topic 4.3

Domagk demonstrates that the red leather dye Prontosil cures streptococcal blood poisoning in mice (and saves his own daughter's arm). Chemists isolate the active ingredient sulfonamide, launching the world's first mass-prescribed class of antibacterial wonder drugs.

Living Timeline Dual-Coding Protocol: Draw visual symbols for Peoria deep fermentation vats, the 1948 NHS charter, Franklin's Photograph 51 double helix, and CT scan slices across the timeline below.

1941–44 • Florey, Chain & Heatley: Penicillin Mass Production

Key Topic 4.4

Howard Florey and Ernst Chain purify penicillin at Oxford, successfully treating mice and Albert Alexander (1941). Heatley and Florey travel to Peoria, Illinois, using corn-steep liquor and deep-tank fermentation to produce 2.3 million doses for D-Day casualties.

1948 • Aneurin Bevan: Foundation of the NHS

Key Topic 4.3

On 5 July 1948, the National Health Service opens at Park Hospital, Manchester. Championed by Bevan and the 1942 Beveridge Report, it nationalises voluntary and municipal hospitals, providing universal healthcare free at the point of clinical delivery.

1953 • Watson, Crick & Franklin: DNA Double Helix Structure

Key Topic 4.1

Using Rosalind Franklin and Maurice Wilkins's X-ray crystallography (Photograph 51), Watson and Crick deduce the double-helix structure of DNA at Cambridge, unlocking hereditary genetic code and modern molecular gene therapy.

1971–2003 • CT Scanners (1971) & The Human Genome Project (2003)

Key Topic 4.2 • 4.1

Godfrey Hounsfield invents the EMI CT scanner, combining computer algorithms with X-rays to image internal organs without surgery. In 2003, the international Human Genome Project maps all 3 billion chemical base pairs in human DNA, founding precision medicine.

Why did the discovery of DNA revolutionize understanding of disease causes, yet take decades to yield treatments?

Core Specification Focus: Genetics and DNA: Mendel; Rosalind Franklin (Photograph 51), Watson and Crick (double helix 1953); the Human Genome Project (1990–2003); genetic screening and hereditary disease.

1 c1900–1940s • Why did doctors assume proteins carried hereditary traits? [1.1]

• Growing wrinkled green garden peas	
• Spotting dark rod shapes inside cells	
• Assuming heavy protein chains hold secrets	

2 1951–1952 • How did Photograph 51 prove DNA's helical structure? [1.2]

• Beaming invisible rays through wet DNA fibers	
• Exposing glass photographic plates for sixty hours	
• Revealing clear fuzzy X-shaped shadow marks	

3 1953 • How did base pairing explain hereditary replication? [1.3]

• Twisting thin metal wire into spirals	
• Locking four chemical pairs like puzzle pieces	
• Rushing to celebratory pints at The Eagle pub	

4 1990–2003 • Why was mapping 3 billion base pairs a medical milestone? [1.4]

• Linking laboratory supercomputers across six nations	
• Reading three billion chemical letters in blood	
• Publishing complete human code free online	

5 2000s–Present • Why did understanding causes outpace finding cures? [1.5]

• Testing saliva drops for faulty breast genes	
• Matching targeted pills to personal body code	
• Struggling to fix broken genetic code inside bodies	

Franklin's 62-hour X-ray exposure revealed the double helix; Watson and Crick celebrated in The Eagle pub.

Question 4: Explain Why [12 marks • 15 mins]

EXTENDED WRITING ASSESSMENT

Explain why there was rapid progress in understanding the causes of disease in the period c1950 to the present. [12 marks]

Stimulus: **Rosalind Franklin's Photograph 51 (1952)** | **The Human Genome Project (1990–2003)** *(You must also use information of your own)*

ACCESSIBLE WRITING FRAMES:

One main reason was that... • For example, ... • This was important because... • Another key factor was... • In particular, ... • Overall, the most decisive reason was... because...

KEY VOCABULARY BANK:

Gregor Mendel • Rosalind Franklin • Photograph 51 (1952) • Watson & Crick • double helix model • Human Genome Project (1990–2003) • gene screening • hereditary disease

Write your response in the space provided below:

(Response continues on facing page)

TIMELINE MISSION Turn to Pages 2–3 (Key Topic 4.1). In the drawing box, sketch Franklin's Photograph 51 cross pattern and Watson & Crick's double helix model. Annotate how base pairing unlocked the causes of hereditary disease! **[ ] Done**

Continue your response to Question 4 below:

• **QUESTION 3: EXPLAIN ONE DIFFERENCE [4 MARKS]** **SCORE: [/ 4] • 5 MINS**

Explain one way in which scientific methods for investigating disease causes in the late twentieth century (c1950–c2000) were different from methods used in the late nineteenth century (c1870–c1900). [4 marks]

Contrast computerized DNA sequencing with manual microscopic staining with chemical dyes.

Were high-tech diagnostic scanners more vital to modern health than state lifestyle campaigns?

Core Specification Focus: Lifestyle factors and diagnosis: shift from infectious to chronic disease; non-invasive diagnosis: Röntgen (X-rays), CT scanners (Hounsfield 1971), MRI; endoscopes, blood and ultrasound testing; lifestyle campaigns.

1 1895–1918 • How did X-rays transform surgical trauma diagnosis in WWI? [2.1]

- **Photographing wife Bertha's hand and gold wedding ring**
- **Driving mobile medical vans to muddy trenches**
- **Finding jagged lead shrapnel without deep cuts**

2 1930s–1960s • Why was fiber-optic endoscopy superior to exploratory surgery? [2.2]

- **Spinning red test tubes in fast centrifuges**
- **Counting abnormal white cells on glass slides**
- **Sliding flexible glass cables down throat tubes**

3 1971–1980s • Why did CT scans transform internal soft-tissue diagnosis? [2.3]

- **Spinning heavy donut scanners around hospital beds**
- **Combining computer pictures into 3D body slices**
- **Using magnetic radio waves to find brain lumps**

4 c1950–1980s • What drove the shift from infectious to lifestyle diseases? [2.4]

- **Banishing deadly childhood fevers with wonder drugs**
- **People living past seventy years of age**
- **Rising chest tumours and clogged heart arteries**

5 1980s–Present • Why do governments favour lifestyle prevention over treatment? [2.5]

- **Clicking compulsory car seatbelts across chests**
- **Extinguishing glowing cigarettes outside pub doors**
- **Taxing sugary fizzy cans and advertising fruit**

Bertha Röntgen cried "I have seen my death!" upon seeing her bones; doctors saw the dawn of non-invasive diagnosis.

Question 5/6: Evaluative Essay [16+4 marks • 20 mins]

EXTENDED WRITING ASSESSMENT

'Technological innovations in medical diagnosis were more important than public health lifestyle campaigns in improving health in Britain c1900–present.' How far do you agree? [16+4 marks]

Stimulus: **Wilhelm Röntgen's X-rays (1895)** | **The 2007 smoking ban in public places** *(You must also use information of your own)*

ACCESSIBLE WRITING FRAMES:

On the one hand, high-tech diagnosis was vital because... • For example, ... • This was significant because... • On the other hand, lifestyle campaigns mattered because... • In particular, ... • Overall, the most decisive factor was... because...

KEY VOCABULARY BANK:

Wilhelm Röntgen (1895) • X-rays • CT scanner (1971) • Godfrey Hounsfield • MRI • non-invasive imaging • 2007 smoking ban • Change4Life

Write your response in the space provided below:

(Response continues on facing page)

TIMELINE MISSION Turn to Pages 2–3 (Key Topic 4.2). In the drawing box, sketch Bertha Röntgen's hand X-ray and Godfrey Hounsfield's circular CT scanner. Annotate how scanning permanently ended blind exploratory surgery! **[ ] Done**

Hounsfield's CT scanner combined rotating X-rays with digital computers to slice through soft body tissue.

Continue your response to Question 5/6 below:

• **QUESTION 3: EXPLAIN ONE DIFFERENCE [4 MARKS]** **SCORE: [/ 4] • 5 MINS**

Explain one way in which methods of diagnosing illness in the late twentieth century (c1970–present) were different from methods used in the nineteenth century (c1800–c1900). [4 marks]

Contrast non-invasive 3D internal scanning and blood labs with external observation and exploratory surgery.

Was the establishment of the NHS in 1948 more transformative than the development of magic bullets?

Core Specification Focus: Magic bullets and healthcare: Ehrlich (Salvarsan 606), Domagk (Prontosil); high-tech treatments: dialysis, hip replacements; Beveridge Report (1942); Bevan and the foundation of the NHS (1948); universal healthcare.

1 1906–1909 • Why was Salvarsan 606 termed the first "magic bullet"? [3.1]

- **Testing hundreds of toxic yellow arsenic mixtures**
- **Hunting chemical bullets that only hit germs**
- **Curing deadly human syphilis with magic bullet**

2 1932–1935 • How did Prontosil prove synthetic drugs could kill microbes? [3.2]

- **Injecting bright red leather dye into mice**
- **Saving sick daughter from arm-amputation death**
- **Discovering active white sulfa medicine powder**

3 1942 • How did WWII unity create momentum for state healthcare? [3.3]

- **Listing Five Giants that crush poor workers**
- **Surviving Nazi air bombs with community spirit**
- **Demanding free medical care for every citizen**

4 1945–1948 • How did Bevan compromise to persuade resistant doctors? [3.4]

- **Offering private bed perks to stubborn doctors**
- **Delivering free spectacles and dentures to families**
- **Abolishing terrifying doctor bills forever**

5 1950s–Present • Why has modern medical success created financial strain? [3.5]

- **Pumping blood through giant artificial kidney filters**
- **Replacing worn hip joints with metal balls**
- **Charging small fees for paper medicine slips**

Ehrlich tested 605 failed arsenic compounds before compound 606 finally cured syphilis in 1909.

Question 5/6: Evaluative Essay [16+4 marks • 20 mins]

EXTENDED WRITING ASSESSMENT

'The establishment of the National Health Service in 1948 was the most significant turning point in medical care in the period c1900–present.' How far do you agree? [16+4 marks]

Stimulus: **The National Health Service (1948)** | **The discovery of Salvarsan 606 (1909)** *(You must also use information of your own)*

ACCESSIBLE WRITING FRAMES:

On the one hand, the NHS was transformative because... • For example, ... • This was significant because... • On the other hand, magic bullets were crucial because... • In particular, ... • Overall, the most decisive factor was... because...

KEY VOCABULARY BANK:

Paul Ehrlich • Salvarsan 606 (1909) • Gerhard Domagk • Prontosil (1932) • William Beveridge (1942) • Aneurin Bevan • NHS (1948) • free at point of delivery

Write your response in the space provided below: *(Response continues on facing page)*

TIMELINE MISSION Turn to Pages 2–3 (Key Topic 4.3). In the drawing box, sketch Ehrlich's test tube of Salvarsan 606 and the 1948 NHS information leaflet. Annotate how free healthcare abolished medical poverty! **[ ] Done**

Continue your response to Question 5/6 below:

• **QUESTION 3: EXPLAIN ONE DIFFERENCE [4 MARKS]** **SCORE: [/ 4] • 5 MINS**

Explain one way in which healthcare access in Britain after 1948 was different from healthcare access in the 1930s. [4 marks]

Contrast free universal healthcare funded by taxation with private fees and voluntary charity hospitals.

Why did penicillin require a global war and American industrial might to become a mass-produced cure?

Core Specification Focus: Case study: The Antibiotic Revolution: Fleming's discovery (1928); Florey and Chain (Oxford purification 1938–40); wartime need and US mass production (Peoria deep-tank fermentation 1941–44); D-Day supply; antibiotic resistance.

1 1928–1929 • Why could Fleming not turn mould into a clinical medicine? [4.1]

• Finding fluffy mould on dirty glass dish	
• Golden germs dissolving in clear circle halos	
• Failing to separate fragile liquid juice	

2 1938–1940 • How did Norman Heatley solve the purification challenge? [4.2]

• Brewing fungus inside hospital bedpans and milk churns	
• Winning American dollar grants during air raids	
• Freezing brown liquid into precious yellow crystals	

3 1940–1941 • Why was Alexander's trial both a success and a tragedy? [4.3]

• Infecting eight mice and keeping treated four alive	
• Scraping rose thorn scratches on policeman's face	
• Running out of drops before patient recovers	

4 1941–1944 • How did US industrial fermentation enable mass production? [4.4]

• Feeding corn beer wash to rotting melon mould	
• Pumping sterile air through giant ten-thousand-gallon vats	
• Brewing million doses for D-Day landing craft	

5 1944–Present • Why did Fleming warn that antibiotic overuse causes resistance? [4.5]

• Spraying white powder directly into torn soldiers' limbs	
• Three scientists sharing golden Nobel prize medal	
• Overusing capsules creating tough hospital superbugs	

Fleming almost washed his petri dish in Lysol, but noticed a clear halo where mould destroyed golden staphylococci.

Question 4: Explain Why [12 marks • 15 mins]

EXTENDED WRITING ASSESSMENT

Explain why penicillin was successfully developed and mass-produced in the period 1928–1945. [12 marks]

Stimulus: **Alexander Fleming's discovery (1928)** | **The impact of the Second World War** *(You must also use information of your own)*

ACCESSIBLE WRITING FRAMES:

One main reason was that... • For example, ... • This was important because... • Another key factor was... • In particular, ... • Overall, the most decisive reason was... because...

KEY VOCABULARY BANK:

Alexander Fleming (1928) • penicillin • Howard Florey • Ernst Chain • Norman Heatley • Albert Alexander (1941) • Peoria fermentation • mass production (1944)

Write your response in the space provided below:

(Response continues on facing page)

TIMELINE MISSION Turn to Pages 2–3 (Key Topic 4.4). In the drawing box, sketch Fleming's contaminated petri dish and an American 10,000-gallon deep-tank vat. Annotate how penicillin saved Allied troops on D-Day! **[ ] Done**

The Oxford team grew penicillin in bedpans; American factories brewed it in 10,000-gallon fermentation vats.

Continue your response to Question 4 below:

• **QUESTION 3: EXPLAIN ONE SIMILARITY [4 MARKS]**

SCORE: [/ 4] • 5 MINS

Explain one way in which Alexander Fleming's discovery of penicillin in 1928 was similar to Edward Jenner's discovery of the smallpox vaccine in 1796. [4 marks]

Focus on both breakthroughs relying on accidental observations of natural organisms protecting against disease.

Why has the modern fight against lung cancer required state coercion rather than laboratory cures?

Core Specification Focus: Case study: Fight against lung cancer: epidemiology of Doll and Hill (1950); diagnosis: chest X-rays, CT, bronchoscopy; treatments: surgery, radiotherapy, chemotherapy, immunotherapy; government prevention: ad bans, 2007 smoking ban, plain packs (2016).

1 1920s–1950 • Why did early doctors fail to link smoking to lung cancer? [5.1]

- **Men coughing black blood into hospital basins**
- **Blaming smelly road tarmac and lorry exhaust fumes**
- **Colourful billboards showing smiling smoking doctors**

2 1950–1954 • How did statistical epidemiology prove the causal link? [5.2]

- **Interviewing sick patients across twenty London wards**
- **Tracking forty thousand real family doctors over years**
- **Proving heavy smokers face fifty-times higher risk**

3 1962–1971 • Why did government action encounter strong tobacco lobbying? [5.3]

- **Banning glowing cigarette commercials on evening television**
- **Printing bold black warning text on cardboard cartons**
- **Sponsoring fast racing cars to keep brands famous**

4 2000s • Why was passive smoking crucial in banning public smoking? [5.4]

- **Breathing toxic secondhand smoke in crowded rooms**
- **Padlocking pub ashtrays and stepping outside into cold**
- **Raising legal counter buying age to eighteen**

5 2010s–Present • Why does low cancer survival prove prevention beats cure? [5.5]

- **Selling plain ugly mud-green packets with graphic photos**
- **Guiding tiny robotic camera pincers between chest ribs**
- **Boosting immune soldier cells to attack hidden tumours**

Doll and Hill proved smoking caused lung cancer in 1950; Doll immediately extinguished his pipe and lived to age 92.

Question 5/6: Evaluative Essay [16+4 marks • 20 mins]

'Government public health legislation was the main reason for progress in combating lung cancer in the period c1950 to the present.' How far do you agree? [16+4 marks]

Stimulus: **Doll and Hill's research (1950)** | **Technological treatments (radiotherapy, surgery, immunotherapy)** *(You must also use information of your own)*

ACCESSIBLE WRITING FRAMES:

On the one hand, government legislation was decisive because... • For example, ... • This was significant because... • On the other hand, high-tech treatments mattered because... • In particular, ... • Overall, the most decisive factor was... because...

KEY VOCABULARY BANK:

Richard Doll • Austin Bradford Hill (1950) • British Doctors Study • 1965 TV ad ban • 2007 public smoking ban • plain packaging (2016) • radiotherapy • immunotherapy

Write your response in the space provided below:

(Response continues on facing page)

TIMELINE MISSION Turn to Pages 2–3 (Key Topic 4.5). In the drawing box, sketch a 1950s doctor cigarette advert and a 2016 olive-green plain pack. Annotate how state legislation slashed smoking from 65% to under 13%! **[ ] Done**

Continue your response to Question 5/6 below:

• **QUESTION 3: EXPLAIN ONE SIMILARITY [4 MARKS]** **SCORE: [/ 4] • 5 MINS**

Explain one way in which government intervention against lung cancer after 1965 was similar to government action in the 1875 Public Health Act. [4 marks]

Focus on both representing state intervention and compulsory legislation replacing laissez-faire.

Compare 1875 sewers with 2007 smoking bans: state compulsion consistently achieves greater public health impact than laissez-faire.

KEY TOPIC 4 NOTES & EXAMINATION STRATEGY

Notes: Record key causal links, essay plans, and teacher feedback for this unit.

PUPIL NAME: _____ TARGET: _____ CURRENT GRADE: _____

ENQUIRY	ASSESSMENT COMPONENT FOCUS	PAGE	SCORE	TEACHER COMMENT & NEXT STEPS
KT4.1	**Q4:** Explain Why Rapid Progress in Causes c1950–Present	p. 5–6	/ 12	
	Q3: Explain One Difference (Investigating Causes: 1880s vs 1990s)	p. 6	/ 4	
KT4.2	**Q5/6:** Evaluative Essay on Diagnostic Scanners vs Lifestyle Campaigns	p. 8–9	/ 20	
	Q3: Explain One Difference (Diagnostic Methods: 19th c vs Late 20th c)	p. 9	/ 4	
KT4.3	**Q5/6:** Evaluative Essay on the NHS as a Medical Turning Point (1948)	p. 11–12	/ 20	
	Q3: Explain One Difference (Healthcare Access: 1930s vs Post-1948)	p. 12	/ 4	
KT4.4	**Q4:** Explain Why Penicillin was Mass-Produced (1928–1945)	p. 14–15	/ 12	
	Q3: Explain One Similarity (Fleming 1928 vs Jenner 1796)	p. 15	/ 4	
KT4.5	**Q5/6:** Evaluative Essay on Government Action & Lung Cancer	p. 17–18	/ 20	
	Q3: Explain One Similarity (Anti-Smoking Laws vs 1875 Public Health Act)	p. 18	/ 4	
TOTAL	**Key Topic 4 Cumulative Examination Assessment Portfolio (10 Tasks)**	p. 4–18	/ 104	**Overall Grade:** _____ **Teacher Target:** _____

• DIGITAL LEARNING & RETRIEVAL HUB • INTERACTIVE LESSONS & QUIZZES

Interactive Textbook • 10 Questions Per Enquiry

KT4.1
DNA & Genetics
LESSON HUB
PRACTICE QUIZ
Score: [/ 10]

KT4.2
Lifestyle & Scans
LESSON HUB
PRACTICE QUIZ
Score: [/ 10]

KT4.3
Magic Bullets & NHS
LESSON HUB
PRACTICE QUIZ
Score: [/ 10]

KT4.4
Penicillin Study
LESSON HUB
PRACTICE QUIZ
Score: [/ 10]

KT4.5
Lung Cancer Study
LESSON HUB
PRACTICE QUIZ
Score: [/ 10]

From DNA to the NHS, state action and laboratory science revolutionized healthcare; revision guarantees your GCSE success.

NAME: _____ CLASS: _____ TEACHER: _____

KEY TOPIC 5 • 6 ENQUIRY HOURS • 1914–1918

The British Sector of the Western Front, 1914–1918

Injuries, Treatment and the Trenches in the British Sector



Primary Visual Evidence: *Stretcher Bearers of the RAMC Lifting a Wounded Man out of a Trench* • Gilbert Rogers (c. 1919)

OFFICIAL EDEXCEL SOURCE ARCHIVE • IWM ART 2485 / RAMC HISTORIC COLLECTION

1. FEATURE DRILLS [4M]
Q1(a) & Q1(b): Two 2-mark questions. Identify 1 specific feature + 1 supporting factual detail (5 mins).

2. SOURCE UTILITY [8M]
Q2(a): 3-step evaluation of Content, Provenance (Nature, Origin, Motive), and Contextual Limitations (10 mins).

3. FOLLOW-UP ENQUIRY [4M]
Q2(b): 4-row tabular enquiry: Detail, Historical Question, Source Type, and Clinical Value (5 mins).

COURSE SPECIFICATION • KEY ENQUIRY SEQUENCE (1914–1918)	LEARNT	REVISED
Key Topic 5.1: The Theatre of War: The British Sector, Trench Geography & Battles *Why were trenches designed in zig-zag traverses, and how did terrain at Ypres, Somme, and Arras affect medical treatment?*	☐	☐
Key Topic 5.2: The Trench Environment: Mud, Vermin & Non-Combat Illnesses *How did frontline conditions cause trench foot, trench fever, and dysentery, and what preventive routines were enforced?*	☐	☐
Key Topic 5.3: Battlefield Trauma: High Explosive Shrapnel, Gas Attacks & Infection *Why did explosive artillery produce gas gangrene, and how did the Brodie steel helmet and respirators reduce fatalities?*	☐	☐
Key Topic 5.4: The Chain of Evacuation: Stretcher Bearers, RAP, Dressing Stations & CCS *How did the RAMC and FANY evacuate wounded soldiers through six stations, and why was triage critical at CCSs?*	☐	☐
Key Topic 5.5: Surgical Breakthroughs: The Thomas Splint, Wound Debridement & Mobile X-Rays *How did the Thomas Splint slash femur mortality from 80% to 20%, and why was Carrel-Dakin irrigation necessary?*	☐	☐
Key Topic 5.6: Lifesaving Innovations: Blood Storage, Brain Surgery & Plastic Reconstruction *How did Oswald Robertson establish the first blood bank at Cambrai, and how did Gillies rebuild shattered faces at Sidcup?*	☐	☐

LIVING TIMELINE • PART 1: FRONTLINE CONDITIONS & EARLY TRAUMA (1914–1915)

Instructions: As you study each enquiry lesson, complete the timeline missions by sketching and annotating in the corresponding Key Topic boxes below.

OCT 1914 • Flanders Mud & The 1st Battle of Ypres

Key Topic 5.1

The British Expeditionary Force (BEF) holds the Ypres Salient. Low-lying Flanders clay creates a dangerously high water table; trenches fill with liquid mud. Shelling smashes all natural drainage canals, leaving men standing waist-deep in water and giving rise to the first epidemics of trench foot and dysentery.

APR 1915 • 2nd Battle of Ypres: The First Poison Gas Attack

Key Topic 5.2 • 5.3

German forces release 160 tons of chlorine gas across a 4-mile front. British troops suffer suffocation and fluid in the lungs; soldiers improvise with urine-soaked pads before chemically impregnated cotton-pad respirators (the Hypo Helmet) are rushed to the front. Lice-borne trench fever breaks out across all divisions.

DEC 1915 • The Thomas Splint & Introduction of the Brodie Helmet

Key Topic 5.3 • 5.5

Robert Jones introduces the Thomas Splint to frontline medical posts, stabilizing compound fractures and reducing femur mortality from 80% to 20%. Simultaneously, the War Office begins issuing John Brodie's stamped steel helmet, reducing penetrating shrapnel head trauma by an estimated 75–80%.

If you think your walk to period 1 is rough, try carrying a 14-stone sergeant through 4 miles of knee-deep Flanders clay.

LIVING TIMELINE • PART 2: MASS OFFENSIVES & CLINICAL INNOVATION (1916–1917)

Instructions: Complete the timeline missions by sketching and annotating in the corresponding Key Topic boxes below.

JULY 1916 • The Battle of the Somme: The Triage Crisis

Key Topic 5.4

The BEF suffers 57,000 casualties on 1 July 1916. Stretcher bearers and Casualty Clearing Stations are overwhelmed; motor ambulances struggle along single rutted roads. As a direct result, the RAMC enforces a strict three-tier triage system (Walking Wounded, Immediate Surgery, Moribund) to prevent total collapse.

APR–NOV 1917 • Arras Underground Hospital & Passchendaele Mud

Key Topic 5.1 • 5.4

At Arras, New Zealand tunnelers excavate Thompson's Cave—a 700-bed underground hospital with operating theatres, running water, and electric lights 20m beneath the chalk. At 3rd Ypres (Passchendaele), artillery destroys all drainage systems, creating knee-deep liquid mud requiring 6 bearers per stretcher.

NOV 1917 • Oswald Robertson's Blood Depot & Gillies Facial Reconstruction

Key Topic 5.6

At the Battle of Cambrai, Captain Oswald Robertson deploys the world's first blood bank, using sodium citrate to stop clotting, glucose to preserve red cells, and ice chests to store Group O blood for 28 days. In Britain, Harold Gillies pioneers plastic facial surgery using tubed pedicles at Queen's Hospital, Sidcup.

How did the physical terrain and trench architecture of the British sector dictate medical evacuation and care?

AUDIO • QUIZ

1 1914–1915 • Why was the low-lying topography of Ypres a tactical and medical nightmare?

• Ypres Salient waterlogged	
• Clay soil drainage destruction	
• Deep mud stretcher crisis	

2 1914–1918 • How did zig-zag traverses limit shrapnel injuries and contain shell explosions?

• Zig-zag traverse construction	
• Duckboards & sump drainage	
• Frontline firing step	

3 1915–1918 • Why did narrow communication trenches create acute bottlenecks for stretchers?

• Three-line trench system	
• Communication trench transit	
• Narrow stretcher passages	

4 Nov 1916–1917 • How did the chalk geology at Arras allow safe, sterile surgical conditions?

• Arras chalk caverns (1916)	
• Thompson's Cave hospital	
• 700 underground beds	

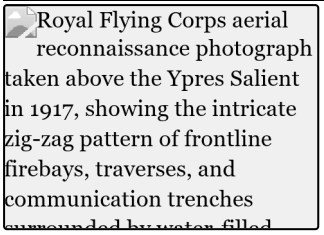
5 1916–1917 • Why did mass-casualty offensives force the RAMC to reorganize evacuation?

• 1916 Somme casualty flood	
• 57,000 first-day casualties	
• 1917 Cambrai tank battle	

Welcome to Flanders: 500 square miles of fermented pig manure, liquid mud, and artillery with terrifying accuracy.

Question 2(a): Source Utility Assessment [8 marks • 10 mins]

SOURCE A • CONTEMPORARY AERIAL RECONNAISSANCE • ROYAL FLYING CORPS **ORIGIN: 1917**


Royal Flying Corps aerial reconnaissance photograph taken above the Ypres Salient in 1917, showing the intricate zig-zag pattern of frontline firebays, traverses, and communication trenches surrounded by water-filled

Royal Flying Corps aerial reconnaissance photograph taken above the Ypres Salient in 1917, showing the intricate zig-zag pattern of frontline firebays, traverses, and communication trenches surrounded by water-filled shell craters.

Historical Provenance: Imperial War Museum / Contemporary Photographic Record (1914–1918).

Study Source A. How useful is Source A for an enquiry into the defensive layout and tactical design of trench systems on the Western Front? Explain your answer, using Source A and your knowledge of the historical context. [8 marks]

Provenance Clue: Consider the nature of an official military aerial reconnaissance photograph taken by the Royal Flying Corps in 1917. What does an overhead view reveal about traverses, fire-steps, and saps, and what ground-level medical and mud conditions does it omit?

ANALYTICAL CONNECTIVES: *Source A is useful for an enquiry into this topic because... • Specifically, the source reveals... • In terms of provenance, considering the author and date... • However, the utility is limited because... • Therefore...*

Write your response to Question 2(a) in the space provided below:

(Response continues on facing page →)

TIMELINE MISSION Turn to Pages 2–3 (Milestone 1). Sketch and annotate the zig-zag traverse pattern and low-lying Ypres water table. **[ ] Done**

Continue your response to Question 2(a) below:

(Concludes on next page →)

Final analytical evaluation & concluding historical verdict for Question 2(a):

• EDEXCEL SECTION A: QUESTION 1 FEATURE DRILLS [4 MARKS • 5 MINS] **2 SEPARATE QUESTIONS • 2 MARKS EACH**

Examiner Mark Scheme: Award 1 mark for identifying a valid historical feature + 1 mark for supporting factual detail. Supporting detail must directly develop the named feature.

1(a) Describe one feature of the system of communication trenches on the Western Front. **[/2]**

FEATURE:

DETAIL:

1(b) Describe one feature of the underground hospital at Arras (Thompson's Cave). **[/2]**

FEATURE:

DETAIL:

Why were environmental pathogens and vermin as dangerous to British troops as enemy artillery?

AUDIO • QUIZ

1 1914–1915 • How did prolonged cold water immersion destroy tissue and produce gangrene?

- Trench foot fungal infection
- Constricted circulation numbness
- Amputation gangrene risk

2 1915–1918 • Why did mandatory officer inspections succeed where voluntary advice had failed?

- Whale oil rubbing
- Dry sock changes
- Daily officer foot inspections

3 1914–1918 • Why did trench fever persist until the body louse insect vector was proven in 1918?

- Trench fever pyrexia
- Body lice transmission
- Delousing bathhouses

4 1914–1918 • How did strict sanitation and chemical water chlorination prevent epidemics?

- Bacterial dysentery illness
- Latrine chloride-of-lime
- Purified water chlorination

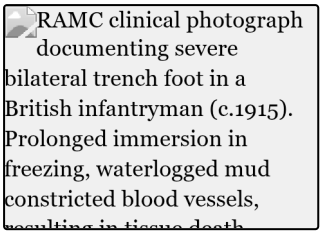
5 1914–1918 • Why was shell shock initially stigmatised as cowardice before being treated medically?

- Shell shock (NYDN)
- Artillery concussive trauma
- Craiglockhart mental convalescence

Whale oil smells like dead fish and regrets, but it beats having your toes amputated by an RAMC surgeon with a bone-saw.

Question 2(b): Follow-Up Enquiry Table [4 marks • 5 mins]

SOURCE A • CONTEMPORARY PHOTOGRAPHIC EVIDENCE • WESTERN FRONT **ORIGIN: July 1916**


RAMC clinical photograph documenting severe bilateral trench foot in a British infantryman (c.1915). Prolonged immersion in freezing, waterlogged mud constricted blood vessels, resulting in tissue death

RAMC clinical photograph documenting severe bilateral trench foot in a British infantryman (c.1915). Prolonged immersion in freezing, waterlogged mud constricted blood vessels, resulting in tissue death, blistering, and wet gangrene.

Historical Provenance: Imperial War Museum / Contemporary Photographic Record (1914–1918).

Study Source A. How could you follow up Source A to find out more about the living conditions and non-combat illnesses experienced by British soldiers in frontline trenches on the Western Front? [4 marks]

Examiner Strategy: Follow up an exact quotation or visual detail. Your question must directly target your chosen detail. Your named source type must be authentic and specific (e.g. RAMC casualty returns, battalion war diaries).

ANALYTICAL CONNECTIVES: *A key detail to follow up is... • The question I would ask is... • The specific historical source type needed is... • This source would help answer my question because...*

Complete the 4-prompt historical enquiry matrix below: *(Response continues on facing page →)*

Detail in Source A that I would follow up:	
Question I would ask:	
Type of source I would use:	
How this would help me find out more:	

Disciplinary Commentary • Clinical Justification of Chosen Source Type:

TIMELINE MISSION Turn to Pages 2–3 (Milestone 2). Draw soldiers applying whale oil and delousing garments in steam ovens. **[ ] Done**

Continue your response to Question 2(b) below:

(Concludes on next page →)

Final analytical evaluation & concluding historical verdict for Question 2(b):

• EDEXCEL SECTION A: QUESTION 1 FEATURE DRILLS [4 MARKS • 5 MINS] **2 SEPARATE QUESTIONS • 2 MARKS EACH**

Examiner Mark Scheme: Award 1 mark for identifying a valid historical feature + 1 mark for supporting factual detail. Supporting detail must directly develop the named feature.

1(a) Describe one feature of the methods used to prevent trench foot on the Western Front. **[/2]**

FEATURE:

DETAIL:

1(b) Describe one feature of the causes of trench fever among British troops. **[/2]**

FEATURE:

DETAIL:

How did high-explosive artillery trauma and chemical poison gas transform surgical treatment?

AUDIO • QUIZ

1 1914–1918 • Why did jagged shrapnel fragments cause far worse infections than rifle bullets?

- Artillery shrapnel (58%)
- High explosive fragments
- Embedded uniform cloth

2 1914–1918 • Why was Clostridium welchii so lethal in deep, airless shrapnel wounds?

- Manured agricultural soil
- Clostridium gas gangrene
- Dead tissue bacteria

3 Late 1915 • Why did the introduction of the Brodie helmet cause an apparent rise in hospital head cases?

- Brodie steel helmet (1915)
- Shrapnel deflection rim
- Fatal head wounds halved

4 Apr 1915–1916 • How did the rapid evolution of chemical respirators counter suffocating gases?

- Chlorine gas Ypres (1915)
- Phosgene suffocation gas
- Urine-soaked pad protection

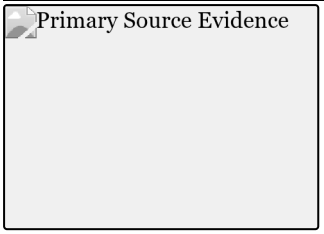
5 1916–1918 • Why was mustard gas far harder to counter than chlorine or phosgene?

- Mustard blistering agent (1917)
- Persistent chemical soil vapour
- Small Box Respirator

The Brodie helmet: Looks like an upside-down soup bowl, but prevents your skull from becoming one.

Question 2(a): Source Utility Assessment [8 marks • 10 mins]

SOURCE B • CONTEMPORARY PHOTOGRAPHIC EVIDENCE • MACHINE GUN CORPS **ORIGIN: July 1916**



Official photographic and archival evidence documenting medical facilities and conditions in the British sector of the Western Front.

Historical Provenance: Imperial War Museum / Contemporary Photographic Record (1914–1918).

Study Source B. How useful is Source B for an enquiry into the methods used to protect British soldiers from poison gas attacks on the Western Front? Explain your answer, using Source B and your knowledge of the historical context. [8 marks]

Provenance Clue: Consider the nature of an authentic British War Office Phenate-Hexamine (PH) anti-gas helmet manufactured in 1915–1916. What physical protective properties does chemically impregnated flannel possess, and what severe combat limitations did it have in battle?

ANALYTICAL CONNECTIVES: *Source B is useful for investigating this topic because... • Specifically, the source provides evidence that... • In terms of provenance, considering the nature and purpose... • However, the source is limited because... • Consequently...*

Write your response to Question 2(a) in the space provided below:

(Response continues on facing page →)

TIMELINE MISSION Turn to Pages 2–3 (Milestone 2). Sketch the Brodie steel helmet and British PH gas hood. **[ ] Done**

Continue your response to Question 2(a) below:

(Concludes on next page →)

Final analytical evaluation & concluding historical verdict for Question 2(a):

• EDEXCEL SECTION A: QUESTION 1 FEATURE DRILLS [4 MARKS • 5 MINS] **2 SEPARATE QUESTIONS • 2 MARKS EACH**

Examiner Mark Scheme: Award 1 mark for identifying a valid historical feature + 1 mark for supporting factual detail. Supporting detail must directly develop the named feature.

1(a) Describe one feature of the effects of poison gas attacks on soldiers on the Western Front. **[/2]**

FEATURE: _____

DETAIL: _____

1(b) Describe one feature of the design of the British Brodie steel helmet. **[/2]**

FEATURE: _____

DETAIL: _____

How successfully did the Chain of Evacuation transport, triage, and treat mass battlefield casualties?

AUDIO • QUIZ

1 1914–1918 • Why was rapid evacuation by stretcher bearers vital to prevent irreversible shock?

• Four-man stretcher teams	
• Regimental Aid Post (RAP)	
• First aid & tourniquets	

2 1914–1918 • What was the tactical role of Dressing Stations in preventing CCS congestion?

• Advanced Dressing Stations (ADS)	
• Field Ambulance RAMC	
• Tetanus anti-toxin serum	

3 1914–1918 • How did the 3-tier triage system maximize survival during mass offensives?

• Casualty Clearing Station (CCS)	
• Three-tier triage sorting	
• Immediate life-saving surgery	

4 1914–1918 • Why were Base Hospitals situated near major coastal rail terminals and ports?

• Coastal Base Hospitals	
• Boulogne & Étaples centres	
• Specialised surgery & convalescence	

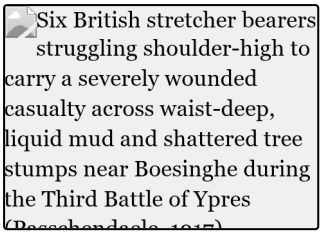
5 1914–1918 • Why were canal barges preferred over trains for head and chest trauma?

• Horse-drawn ambulance delays	
• Motor ambulances introduced	
• Canal barges head-trauma	

The Regimental Aid Post: 200 yards from the German front line, lit by candle-ends, and smelling entirely of iodine and panic.

Question 2(b): Follow-Up Enquiry Table [4 marks • 5 mins]

SOURCE A • CONTEMPORARY PHOTOGRAPHIC EVIDENCE • WESTERN FRONT **ORIGIN: August 1917**



Six British stretcher bearers struggling shoulder-high to carry a severely wounded casualty across waist-deep, liquid mud and shattered tree stumps near Boesinghe during the Third Battle of Ypres (Passchendaele, 1917).

Historical Provenance: Imperial War Museum / Contemporary Photographic Record (1914–1918).

Study Source A. How could you follow up Source A to find out more about the difficulties stretcher bearers faced evacuating casualties from the Western Front? [4 marks]

Examiner Strategy: Follow up an exact quotation or visual detail. Your question must directly target your chosen detail. Your named source type must be authentic and specific (e.g. RAMC casualty returns, battalion war diaries).

ANALYTICAL CONNECTIVES: *A key detail to follow up is... • The question I would ask is... • The specific historical source type needed is... • This source would help answer my question because...*

Complete the 4-prompt historical enquiry matrix below: *(Response continues on facing page →)*

Detail in Source A that I would follow up:	
Question I would ask:	
Type of source I would use:	
How this would help me find out more:	

Disciplinary Commentary • Clinical Justification of Chosen Source Type:

TIMELINE MISSION Turn to Pages 2–3 (Milestone 5). Draw the 6 sequential stages of the Chain of Evacuation. **[ ] Done**

Continue your response to Question 2(b) below:

(Concludes on next page →)

Final analytical evaluation & concluding historical verdict for Question 2(b):

• EDEXCEL SECTION A: QUESTION 1 FEATURE DRILLS [4 MARKS • 5 MINS] **2 SEPARATE QUESTIONS • 2 MARKS EACH**

Examiner Mark Scheme: Award 1 mark for identifying a valid historical feature + 1 mark for supporting factual detail. Supporting detail must directly develop the named feature.

1(a) Describe one feature of the triage system used at Casualty Clearing Stations on the Western Front. **[/2]**

FEATURE:

DETAIL:

1(b) Describe one feature of the work of the First Aid Nursing Yeomanry (FANY) on the Western Front. **[/2]**

FEATURE:

DETAIL:

Why were the Thomas Splint, debridement, and mobile X-rays decisive surgical breakthroughs?

AUDIO • QUIZ

1 1914–1918 • Why was surgical excision of dead tissue more effective than chemical antiseptics?

• Wound surgical debridement	
• Dead tissue excision	
• Delayed primary closure	

2 1915–1918 • How did continuous chemical irrigation solve the problem of deep wound infection?

• Carrel-Dakin antiseptic irrigation	
• Sodium hypochlorite bleach	
• Rubber perfusion tubes	

3 Dec 1915–1916 • How did rigid traction prevent fatal haemorrhage and shock from broken femurs?

• Thomas splint (1915)	
• Compound femur traction	
• Mortality reduced (80% to 20%)	

4 1914–1918 • Why was locating shrapnel on an X-ray vital before operating on gas gangrene?

• Mobile radiographic vans	
• Van dynamo generators	
• Locating embedded shrapnel	

5 1914–1918 • Why did undetected cloth fragments remain the primary cause of post-op gangrene?

• Fragile vacuum tubes	
• Overheating radiation limits	
• Undetected cloth fragments	

The Thomas Splint: Before Robert Jones introduced it, an 80% chance of death; after Jones, an 80% chance of living to complain about the food.

Question 2(a): Source Utility Assessment [8 marks • 10 mins]

SOURCE A • CONTEMPORARY MEDICAL ARTIFACT • STANDARD ISSUE **ORIGIN: c.1916**



Operating theatre suite inside No. 23 Casualty Clearing Station on the Western Front. Multiple operating tables with surgical instruments and overhead illumination enabled specialist surgical teams to perform urgent abdominal laparotomies and debridement simultaneously.

Historical Provenance: Imperial War Museum / Contemporary Photographic Record (1914–1918).

Study Source A. How useful is Source A for an enquiry into the methods used to treat wounded soldiers with fractured limbs on the Western Front? Explain your answer, using Source A and your knowledge of the historical context. [8 marks]

Provenance Clue: Consider the nature of an authentic instructional photograph taken in an RAMC military hospital training ward c.1916. What does it demonstrate about mechanical traction and splinting, and why might an instructional training photograph differ from emergency frontline conditions in a mud dugout?

ANALYTICAL CONNECTIVES: *Source A is useful for an enquiry into this topic because... • Specifically, the source demonstrates that... • In terms of provenance, considering the contemporary context... • However, the utility is limited because... • Therefore...*

Write your response to Question 2(a) in the space provided below: *(Response continues on facing page →)*

TIMELINE MISSION Turn to Pages 2–3 (Milestone 3). Sketch the Thomas Splint groin ring and traction cord. **[ ] Done**

Continue your response to Question 2(a) below:

(Concludes on next page →)

Final analytical evaluation & concluding historical verdict for Question 2(a):

• EDEXCEL SECTION A: QUESTION 1 FEATURE DRILLS [4 MARKS • 5 MINS] **2 SEPARATE QUESTIONS • 2 MARKS EACH**

Examiner Mark Scheme: Award 1 mark for identifying a valid historical feature + 1 mark for supporting factual detail. Supporting detail must directly develop the named feature.

1(a) Describe one feature of the Thomas Splint used on the Western Front. **[/2]**

FEATURE: _____

DETAIL: _____

1(b) Describe one feature of the Carrel-Dakin method of treating infected wounds. **[/2]**

FEATURE: _____

DETAIL: _____

How did stored blood banks, specialized neurosurgery, and plastic reconstruction transform wartime survival?

AUDIO • QUIZ

1 1914–1915 • How did sodium citrate liberate blood transfusion from donor-to-patient tubing?

• Richard Lewisohn sodium citrate	
• Uncoupled indirect transfusion	
• Preventing blood clotting	

2 1916–1917 • Why was glucose addition essential to prevent red blood cell breakdown?

• Rous & Turner citrate-glucose	
• Stored refrigerated blood	
• Four-week viability window	

3 Nov 1917 • Why did pre-stored blood transform surgical survival during surprise assaults?

• Oswald Robertson (1917)	
• Cambrai blood depot	
• Type O universal donors	

4 1917–1918 • Why was local anaesthesia superior to general anaesthetic in head trauma?

• Harvey Cushing neurosurgery	
• Local anaesthetic brain operations	
• Magnet shrapnel extraction	

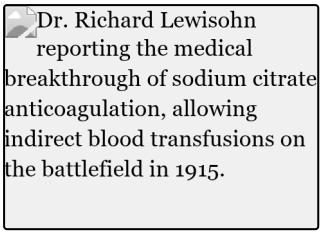
5 1917–1918 • How did the tubed pedicle technique prevent skin grafts from rotting and dying?

• Harold Gillies Sidcup (1917)	
• Queen's Hospital facial surgery	
• Tubed pedicle skin grafting	

Robertson's Blood Depot: Ice chests, sodium citrate, and refrigerated blood at Cambrai—proof that cold beer isn't the only thing worth chilling.

Question 2(b): Follow-Up Enquiry Table [4 marks • 5 mins]

SOURCE A • PRIMARY MEDICAL ARTIFACT • CAMBRAI BLOOD DEPOT **ORIGIN: 1917**



Dr. Richard Lewisohn reporting the medical breakthrough of sodium citrate anticoagulation, allowing indirect blood transfusions on the battlefield in 1915.

Dr. Richard Lewisohn reporting the medical breakthrough of sodium citrate anticoagulation, allowing indirect blood transfusions on the battlefield in 1915.

Historical Provenance: Annals of Surgery, Vol. 62, New York, July 1915.

Study Source A. How could you follow up Source A to find out more about the success of stored blood transfusions at the Battle of Cambrai in 1917? [4 marks]

Examiner Strategy: Follow up an exact quotation or visual detail. Your question must directly target your chosen detail. Your named source type must be authentic and specific (e.g. RAMC casualty returns, battalion war diaries).

ANALYTICAL CONNECTIVES: *A key detail to follow up is... • The question I would ask is... • The specific historical source type needed is... • This source would help answer my question because...*

Complete the 4-prompt historical enquiry matrix below: *(Response continues on facing page →)*

Detail in Source A that I would follow up:	
Question I would ask:	
Type of source I would use:	
How this would help me find out more:	

Disciplinary Commentary • Clinical Justification of Chosen Source Type:

TIMELINE MISSION Turn to Pages 2–3 (Milestone 6). Sketch Oswald Robertson's iced blood chest and Harold Gillies' tubed pedicle. **[ ] Done**

Continue your response to Question 2(b) below:

(Concludes on next page →)

Final analytical evaluation & concluding historical verdict for Question 2(b):

• EDEXCEL SECTION A: QUESTION 1 FEATURE DRILLS [4 MARKS • 5 MINS] **2 SEPARATE QUESTIONS • 2 MARKS EACH**

Examiner Mark Scheme: Award 1 mark for identifying a valid historical feature + 1 mark for supporting factual detail. Supporting detail must directly develop the named feature.

1(a) Describe one feature of Harvey Cushing's surgical techniques in operating on head wounds on the Western Front. **[/2]**

FEATURE:

DETAIL:

1(b) Describe one feature of the surgical methods used by Harold Gillies at Queen's Hospital, Sidcup. **[/2]**

FEATURE:

DETAIL:

STUDENT ASSESSMENT RECORD • KEY TOPIC 5 TRACKER

The British Sector of the Western Front, 1914–1918: Injuries, Treatment and the Trenches

PUPIL: _____ **TARGET GRADE:** ☐ **PREDICTED:** ☐ **ATTITUDE:** 1 • 2 • 3 • 4 • 5

#	ENQUIRY / HISTORIC ENVIRONMENT FOCUS	DO NOW (10M)	Q1 FEATURES (4M)	Q2 SOURCE ENQUIRY	LESSON TOTAL
1	**KT5.1: The Theatre of War: The British Sector, Trench Geography & Battles** *"How did the physical terrain and trench architecture of the British sector dictate medical evacuation and care?"*	/ 10	/ 4	Q2(a) Utility: / 8	/ 22
2	**KT5.2: The Trench Environment: Mud, Vermin & Non-Combat Illnesses** *"Why were environmental pathogens and vermin as dangerous to British troops as enemy artillery?"*	/ 10	/ 4	Q2(b) Follow-up: / 4	/ 18
3	**KT5.3: Battlefield Trauma: High Explosive Shrapnel, Gas Attacks & Infection** *"How did high-explosive artillery trauma and chemical poison gas transform surgical treatment?"*	/ 10	/ 4	Q2(a) Utility: / 8	/ 22
4	**KT5.4: The Chain of Evacuation: Stretcher Bearers, RAP, Dressing Stations & CCS** *"How successfully did the Chain of Evacuation transport, triage, and treat mass battlefield casualties?"*	/ 10	/ 4	Q2(b) Follow-up: / 4	/ 18
5	**KT5.5: Surgical Breakthroughs: The Thomas Splint, Wound Debridement & Mobile X-Rays** *"Why were the Thomas Splint, debridement, and mobile X-rays decisive surgical breakthroughs?"*	/ 10	/ 4	Q2(a) Utility: / 8	/ 22
6	**KT5.6: Lifesaving Innovations: Blood Storage, Brain Surgery & Plastic Reconstruction** *"How did stored blood banks, specialized neurosurgery, and plastic reconstruction transform wartime survival?"*	/ 10	/ 4	Q2(b) Follow-up: / 4	/ 18
KEY TOPIC 5 CUMULATIVE ASSESSMENT TOTALS		**Do Now: / 60**	**Q1 Feat: / 24**	**Q2 Enq: / 36**	**/ 120**

TEACHER FORMATIVE ASSESSMENT • WWW / EBI FEEDBACK

WHAT WENT WELL (WWW):

_____

_____

_____

EVEN BETTER IF (EBI):

_____

_____

_____

Teacher Signature: _____ **Date:** _____

DIGITAL QUIZZING HUB • 20-QUESTION PRACTICE QUIZZES

Scan with smartphone camera to open live interactive 20-question self-marking quizzes

KT 5.1	KT 5.2	KT 5.3	KT 5.4	KT 5.5	KT 5.6
SCAN TO QUIZ	SCAN TO QUIZ	SCAN TO QUIZ	SCAN TO QUIZ	SCAN TO QUIZ	SCAN TO QUIZ
Score: / 20	**Score: / 20**	**Score: / 20**	**Score: / 20**	**Score: / 20**	**Score: / 20**

Retain this student assessment record for parent consultations and GCSE revision monitoring.